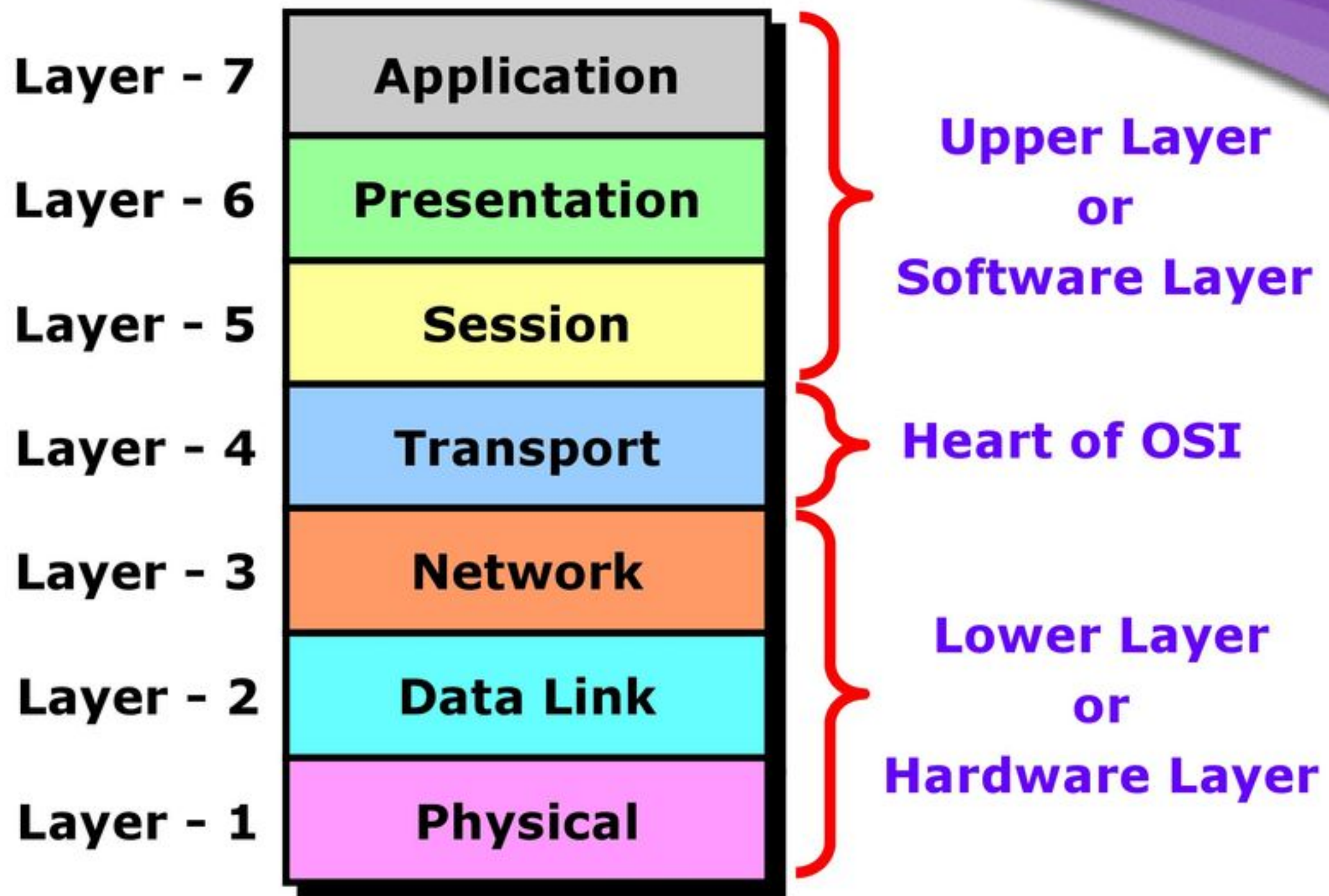
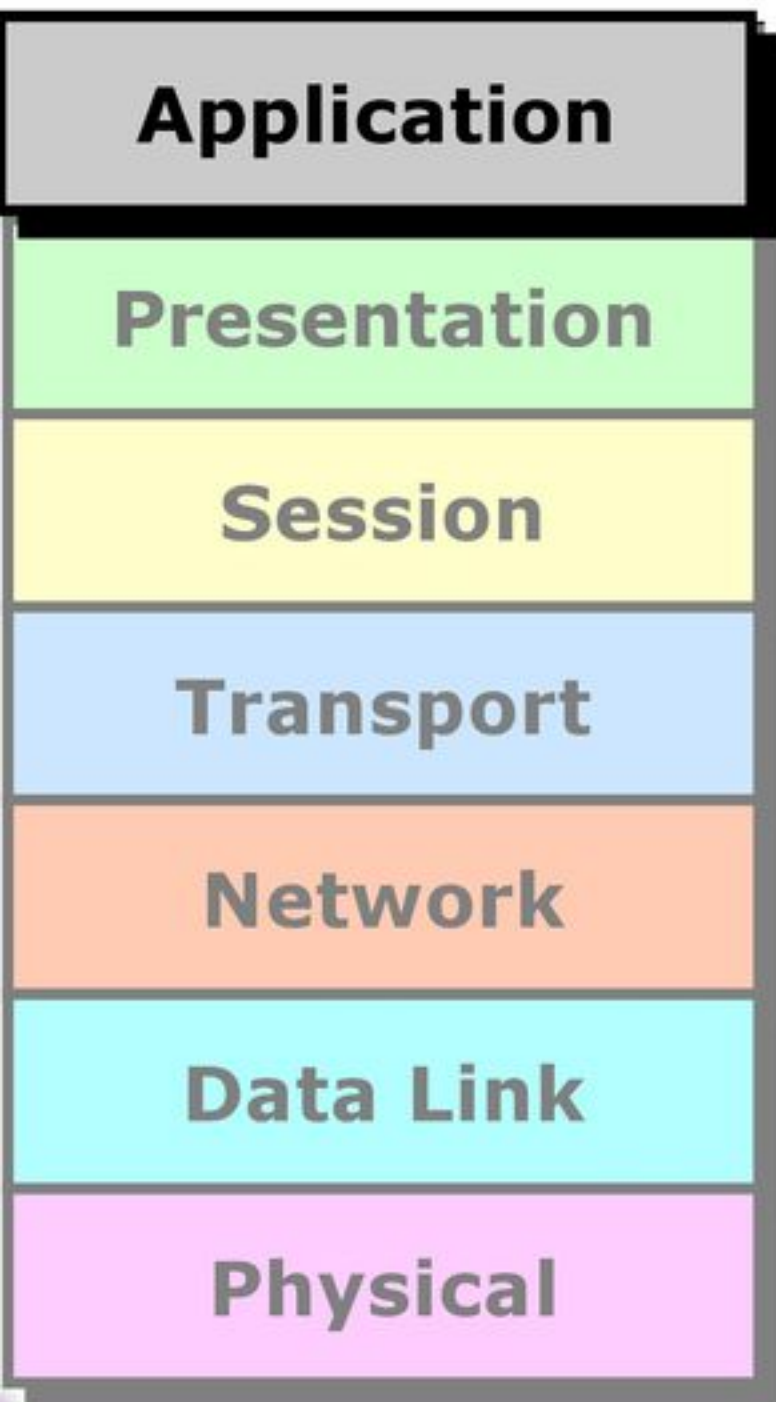


OSI LAYERS

- **OSI means Open System Interconnect model.**
- **Developed by the International Organization for Standardization in 1974.**
- **It consists of seven layers.**
- **Each layer has a different but specific processing function.**





Application Layer is responsible for providing Networking Services to the user. It is also known as Desktop Layer. Identification of Services is done using Port Numbers.

Ports are Entry and Exit Points to the Layer

Total No. Ports	0 – 65535
Reserved Ports	0 – 1023
Open Client Ports	1024 – 65535



Action canceled

Internet Explorer was unable to link to the Web page you requested. The page might be temporarily unavailable.

Please try the following:

- Click the [Refresh](#) button, or try again later.
- If you have visited this page previously and you want to view what has been stored on your computer, click **File**, and then click **Work Offline**.
- For information about offline browsing with Internet Explorer, click the **Help** menu, and then click **Contents and Index**.

Internet Explorer

File Edit View Favorites Tools Help

Back Forward Stop Search Folders History Print Copy Paste Close Reload Home

Address ftp://ftp.microsoft.com/ Go Links >>

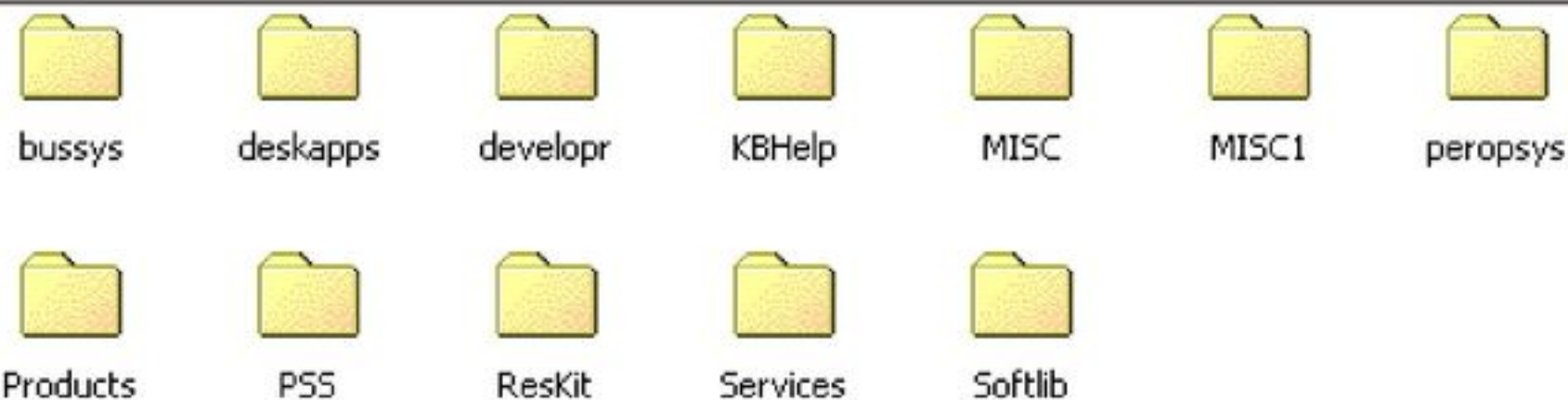


ftp.microsoft.com

Server: ftp.microsoft.com
User Name: Anonymous

This is FTP.Microsoft.Com.

[Click here](#) to learn about browsing FTP sites.



User: Anonymous

Internet

Examples of Networking Services

Service

Port No.

HTTP

80

FTP

21

SMTP

25

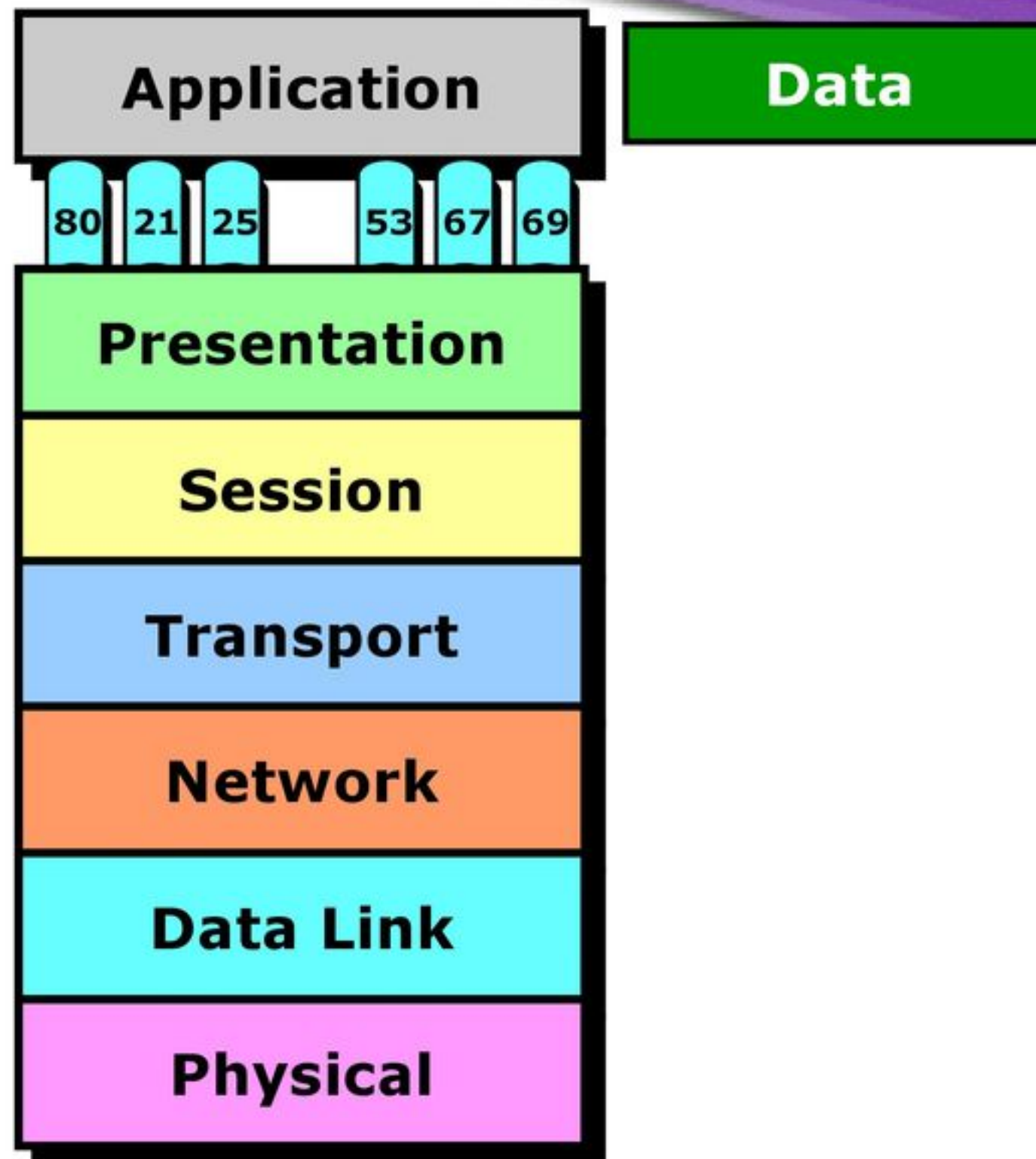
TELNET

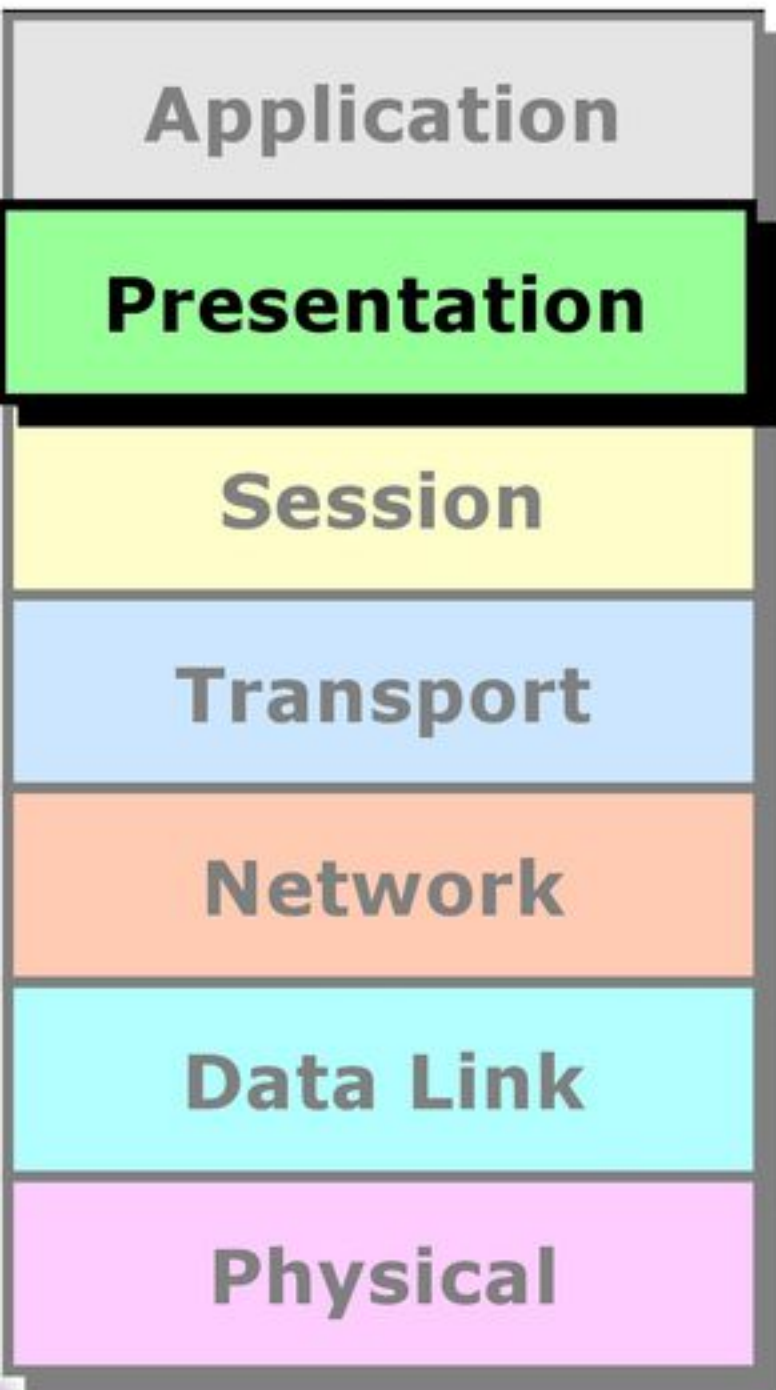
23

TFTP

69







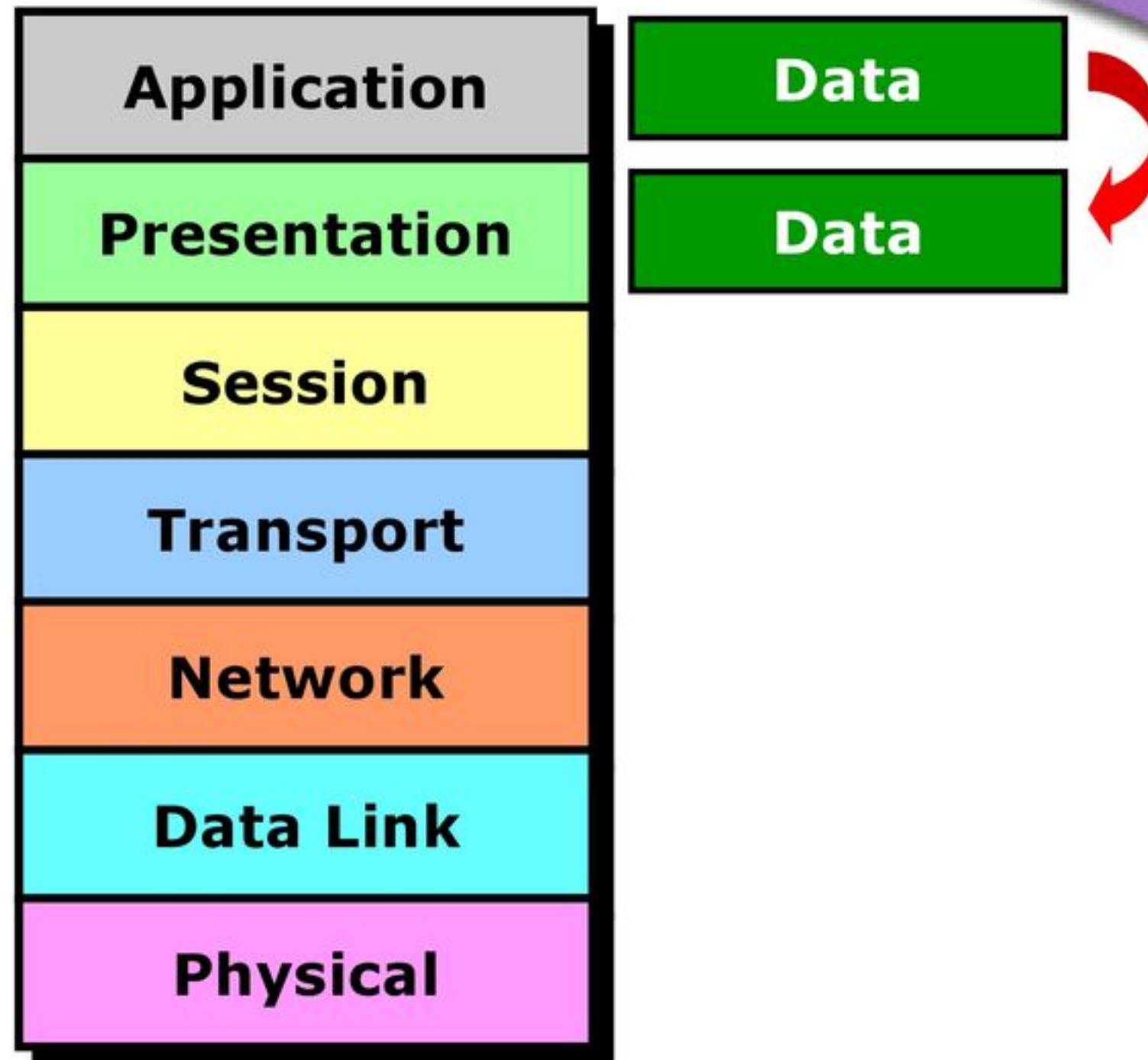
Presentation Layer is responsible for converting data into standard format.

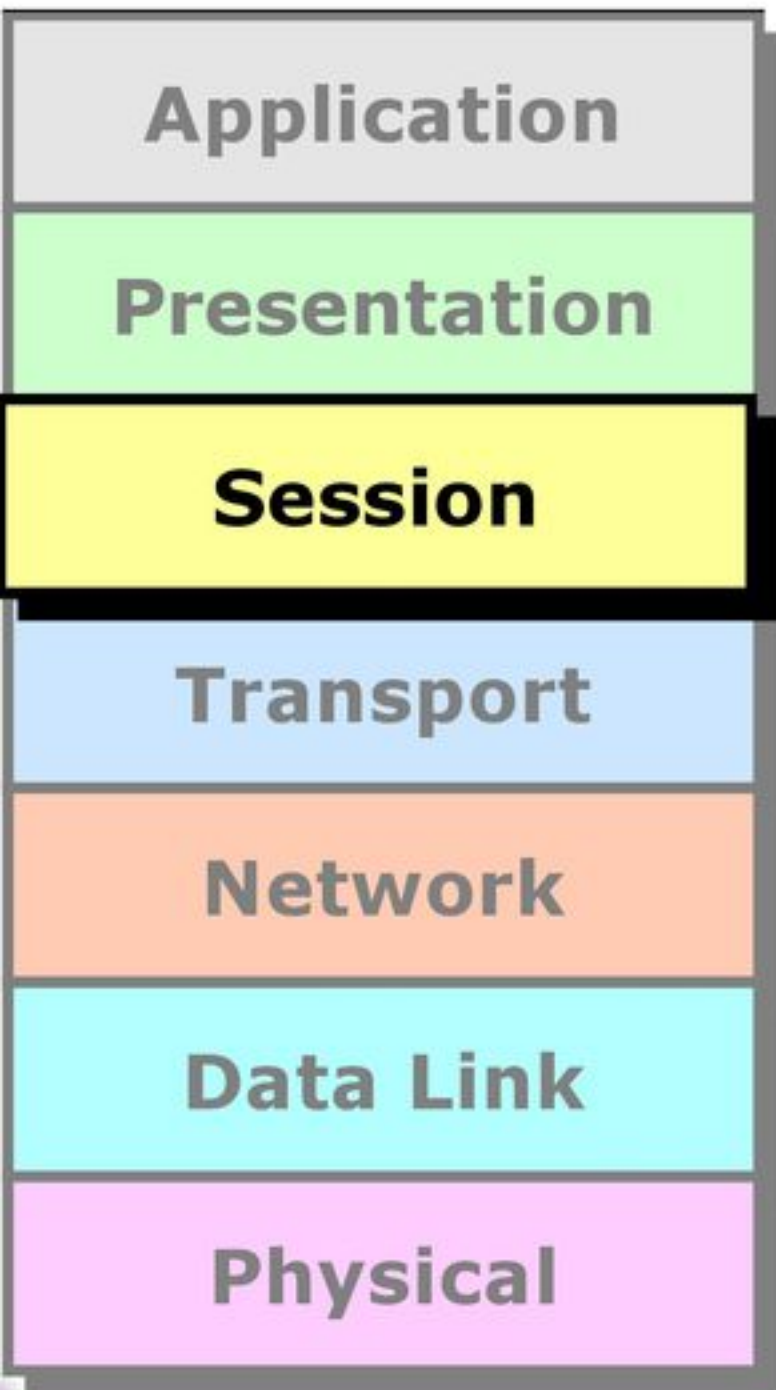
Examples : ASCII, EBCDIC, JPEG, MPEG, BMP, MIDI, WAV, MP3

Following tasks are perform at Presentation layer :

Encoding – Decoding
Encryption – Decryption
Compression – Decompression







Session Layer is responsible for establishing, maintaining and terminating session.
Session ID works at Session Layer.

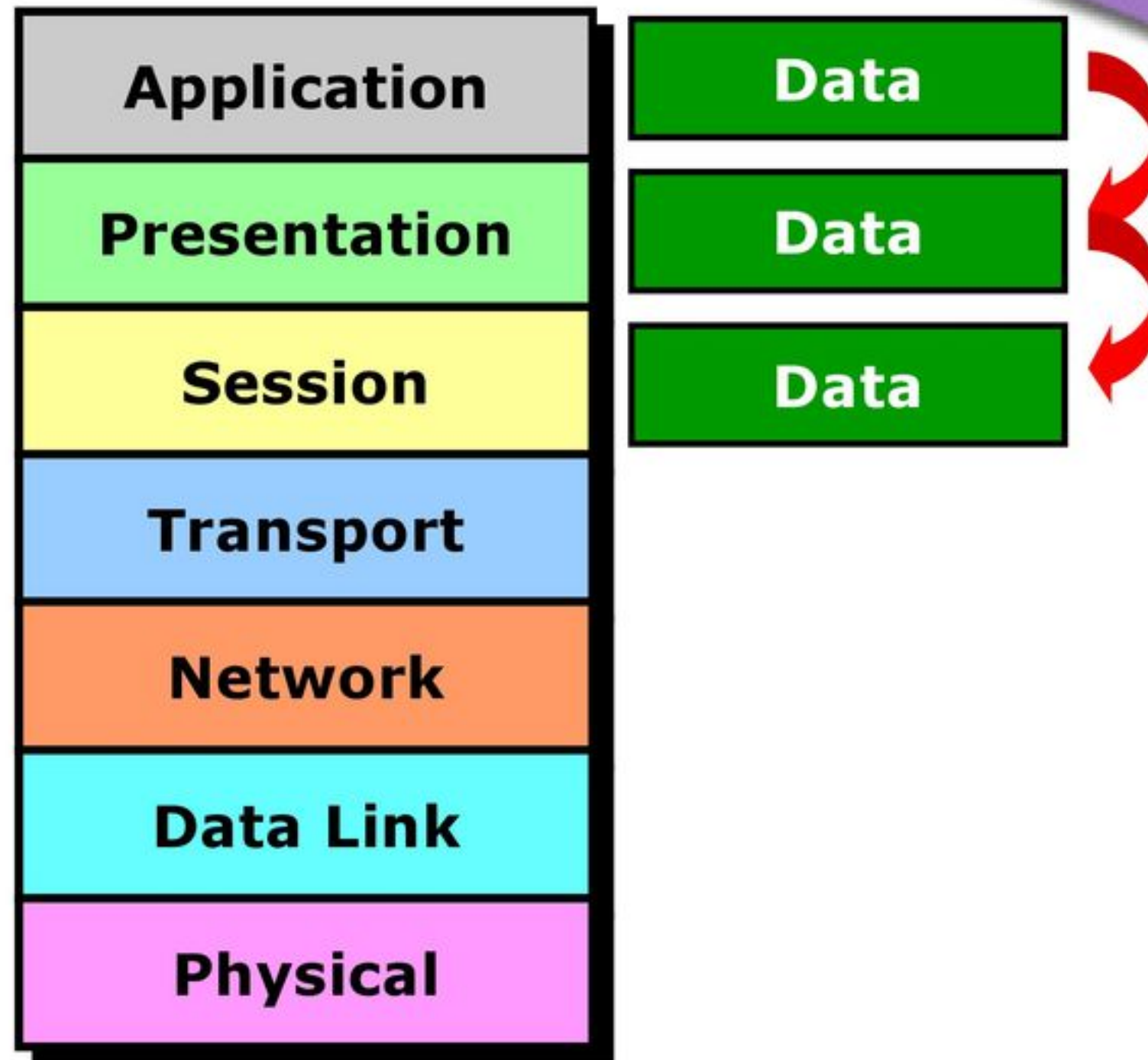
Examples :

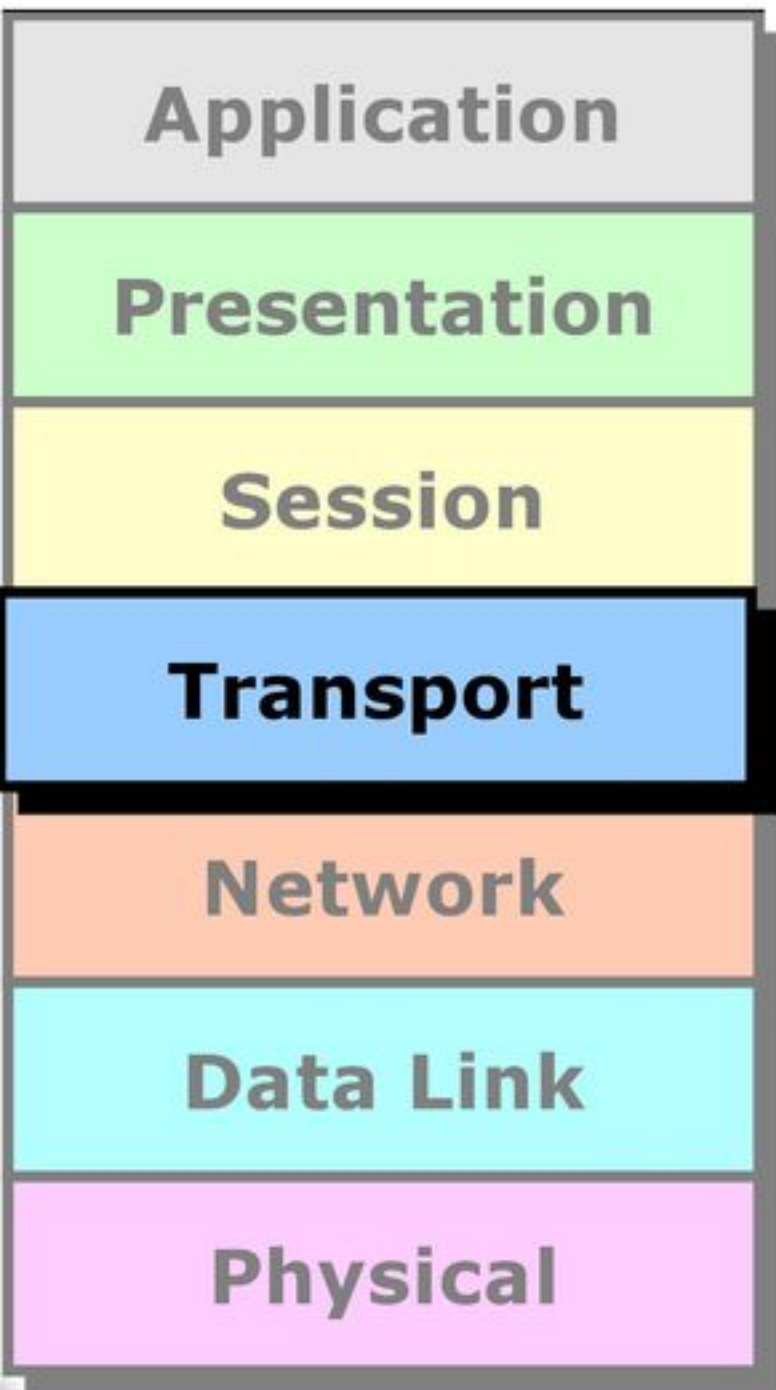


RPC → Remote Procedure Call

SQL → Structured Query Language

NFS → Network File System





Transport Layer is responsible for end-to-end connectivity. It is also known as the heart of OSI Layers. Following tasks are performed at the Transport Layer : -

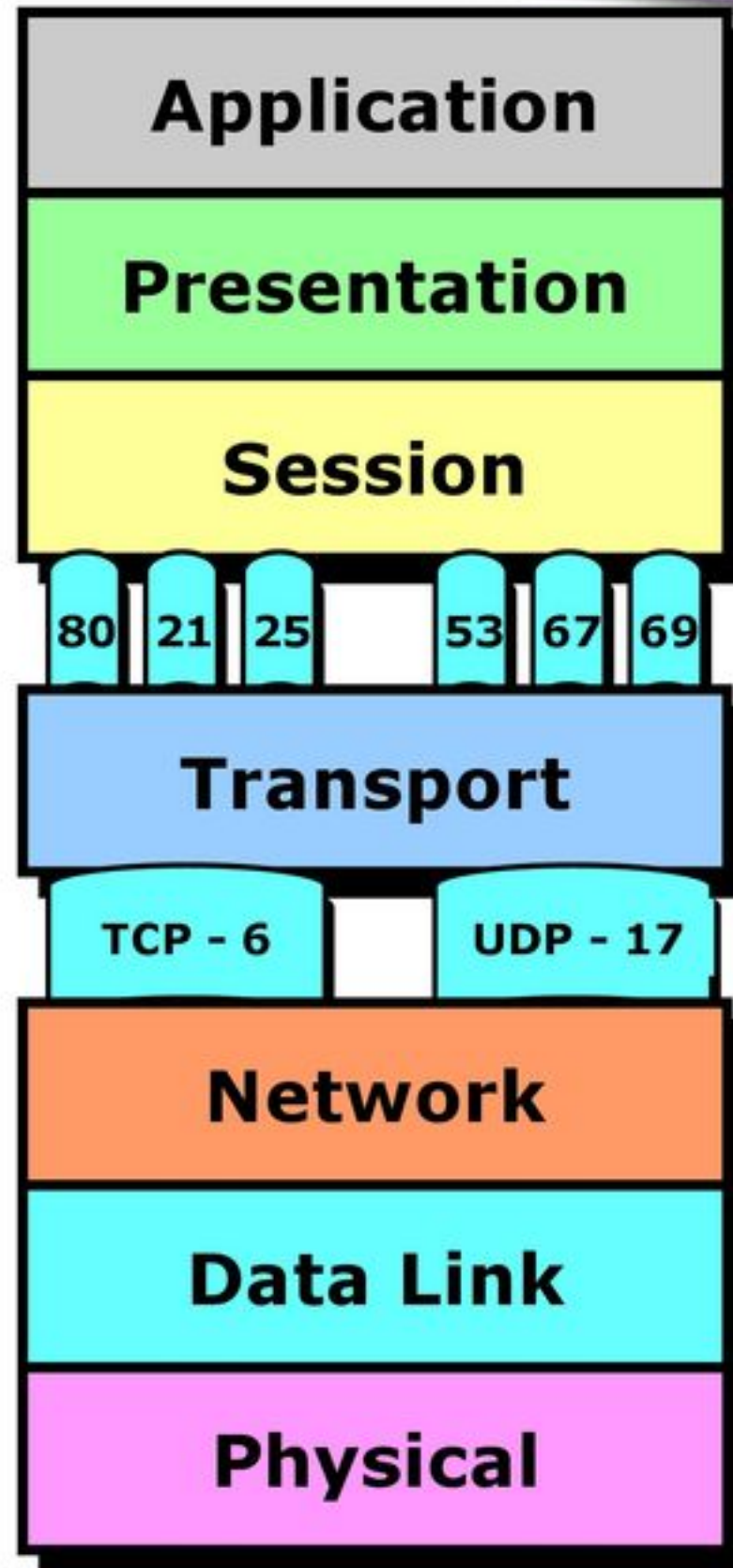
- **Identifying Service**
- **Multiplexing & De-multiplexing**
- **Segmentation**
- **Sequencing & Reassembling**
- **Error Correction**
- **Flow Control**

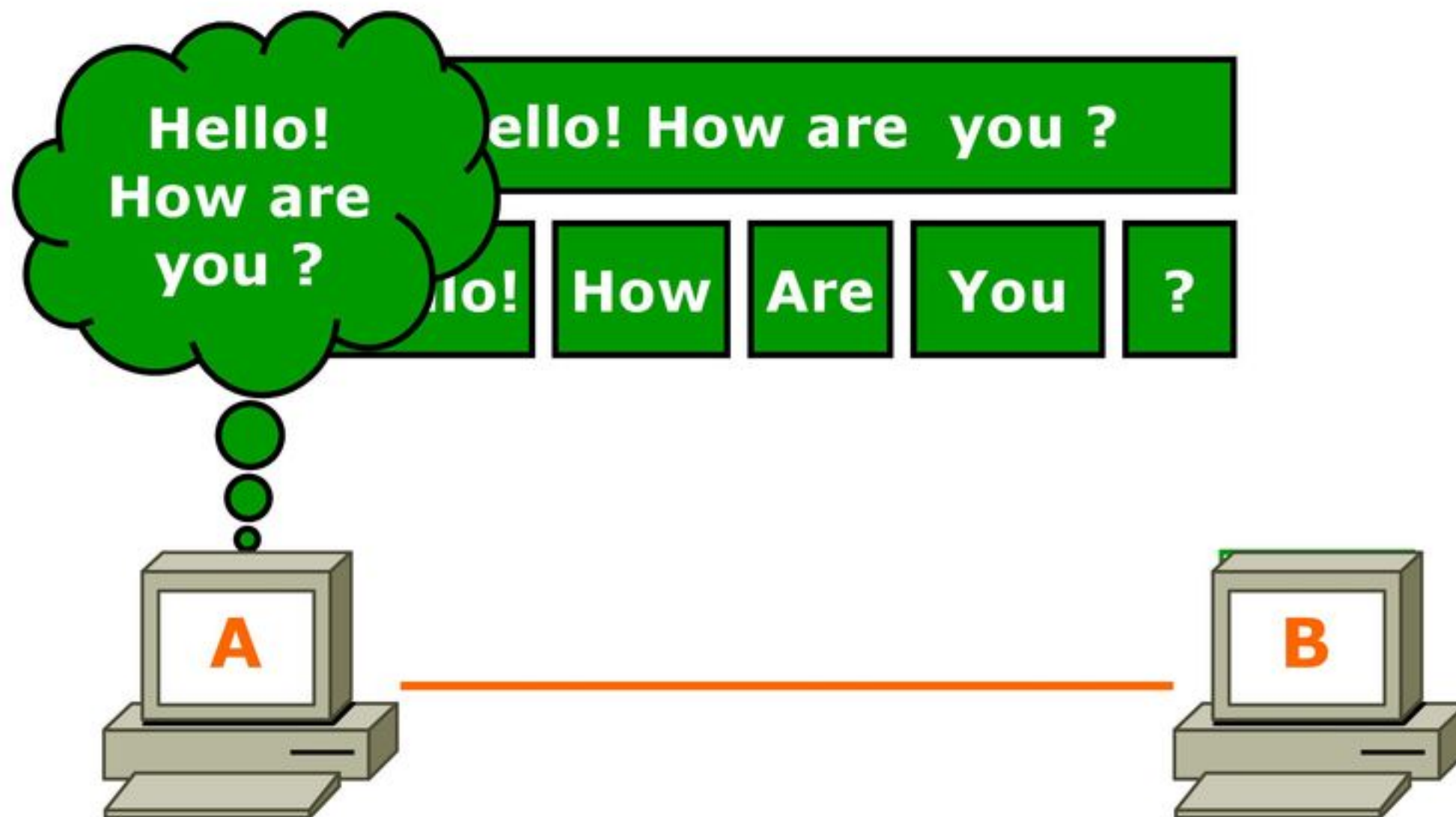
TCP

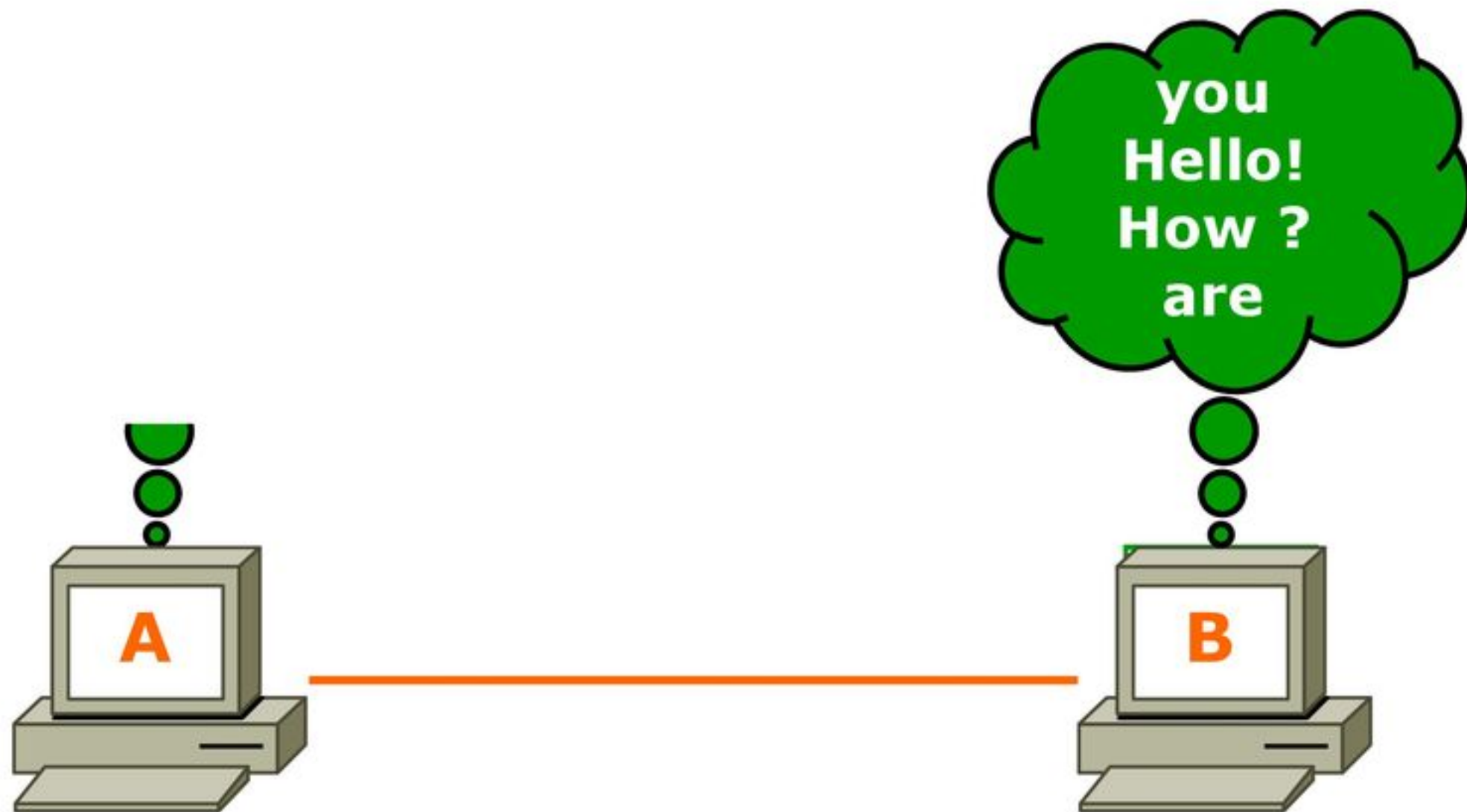
- **Transmission Control Protocol**
- **Connection Oriented**
- **Acknowledgement**
- **Reliable**
- **Slower**
- **Port No. 6**
- **e.g. HTTP, FTP, SMTP**

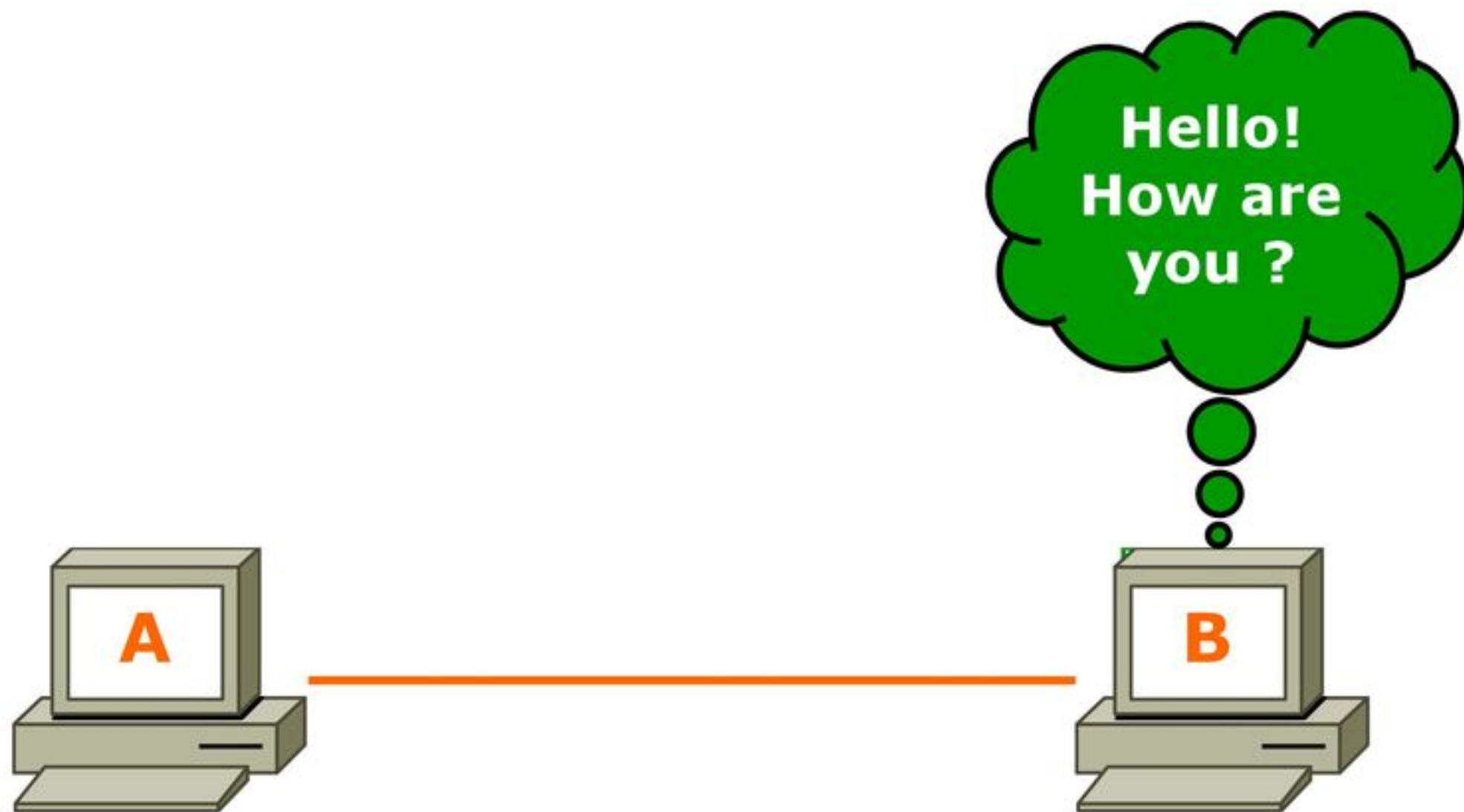
UDP

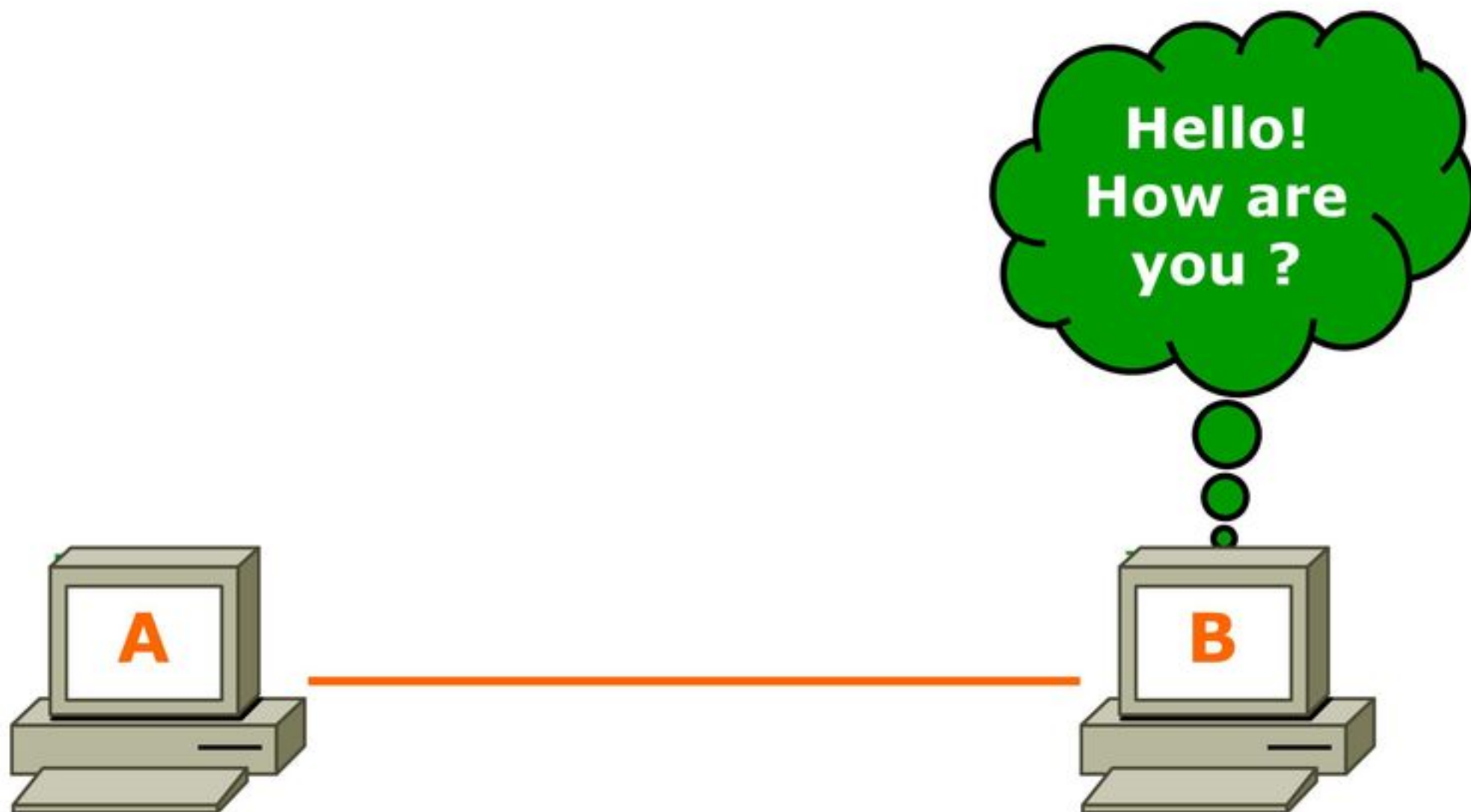
- **User Datagram Protocol**
- **Connection Less**
- **No Acknowledgement**
- **Unreliable**
- **Faster**
- **Port No. 17**
- **e.g. DNS, DHCP, TFTP**

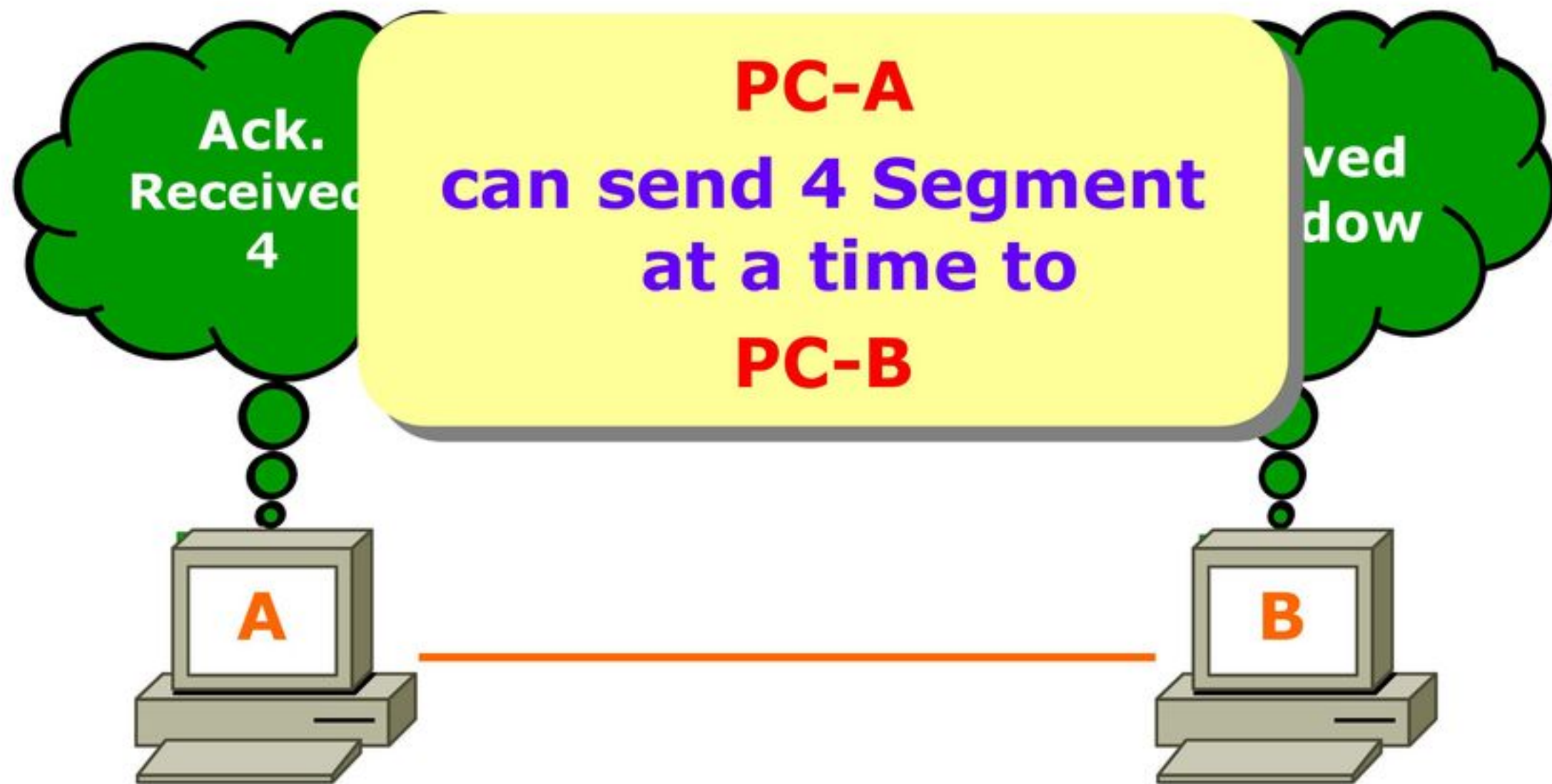


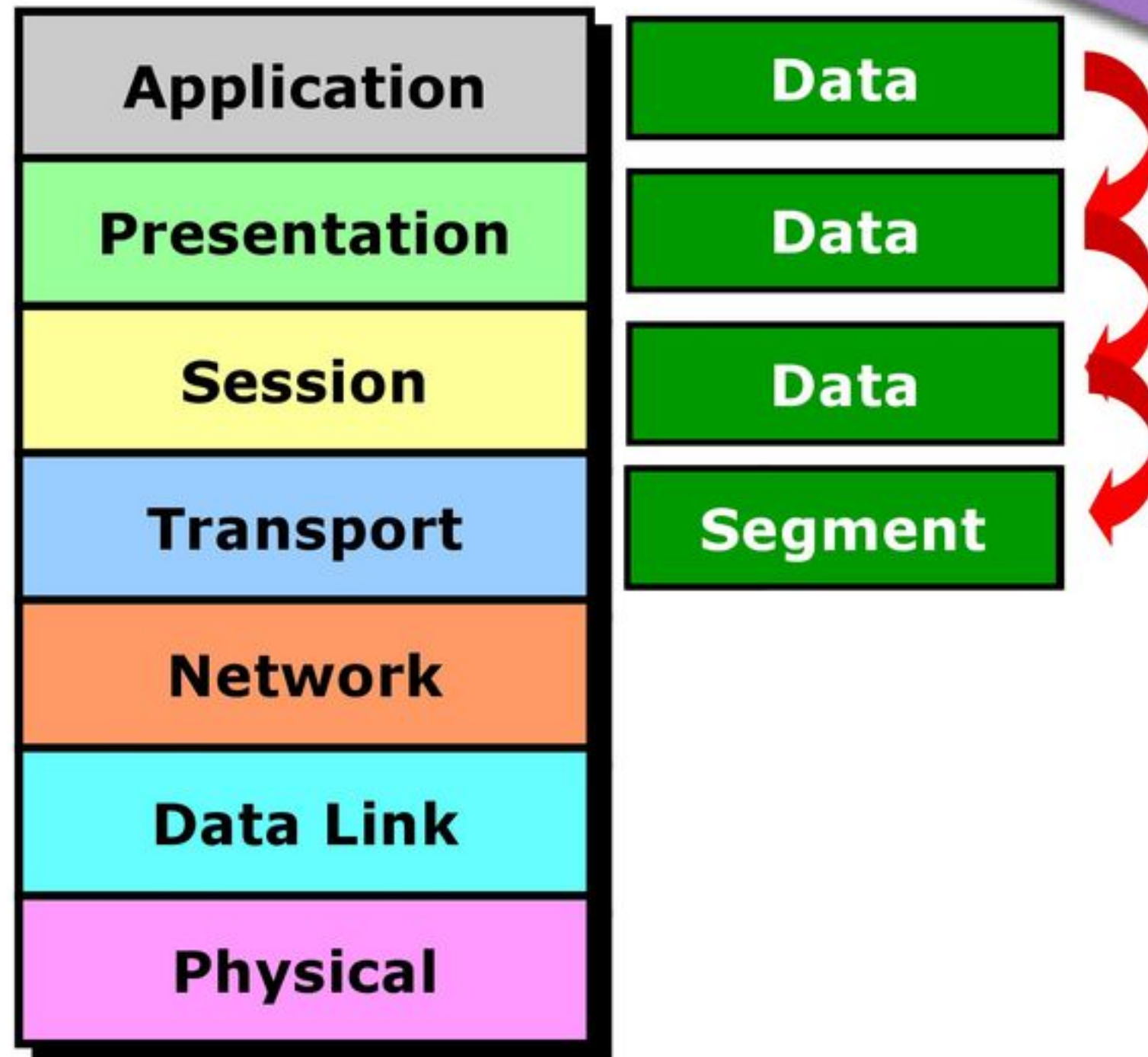


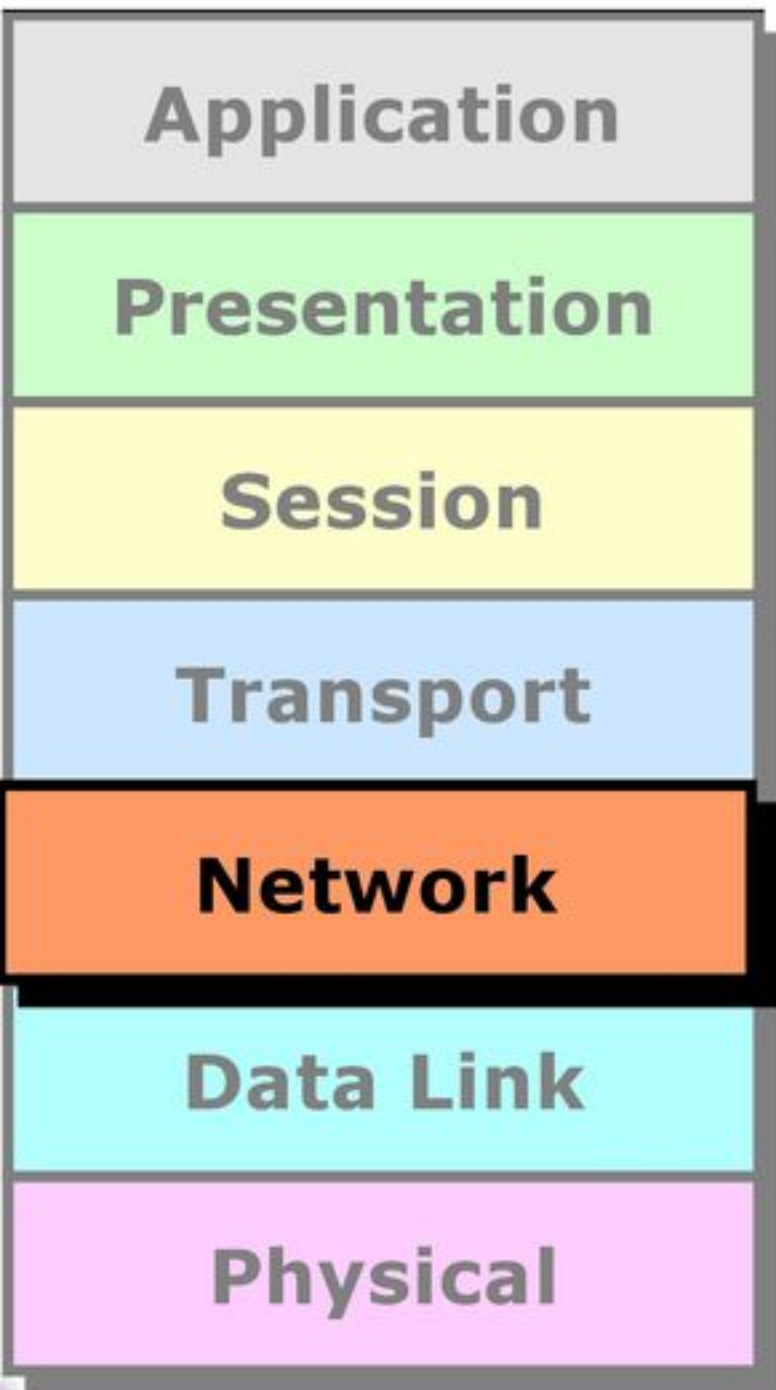








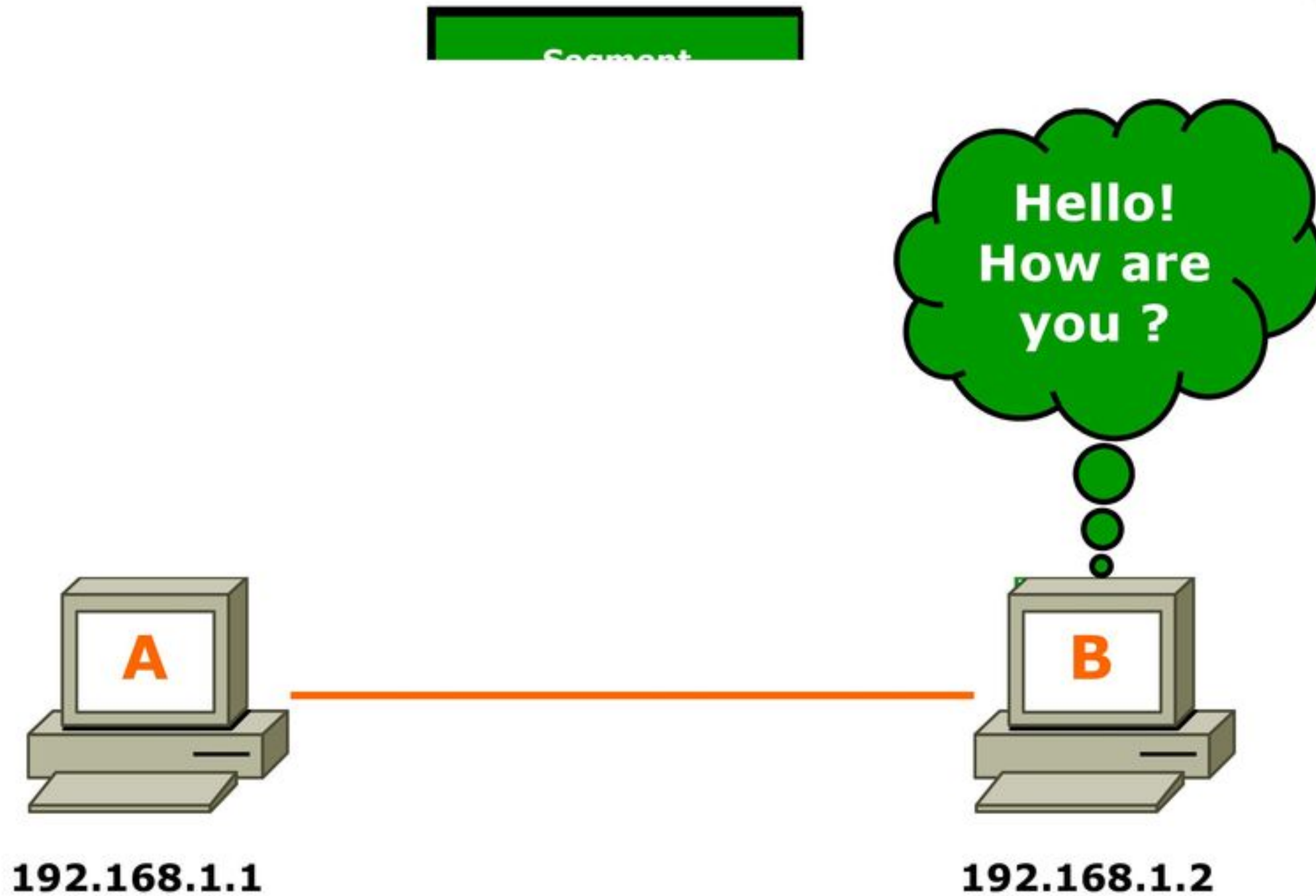


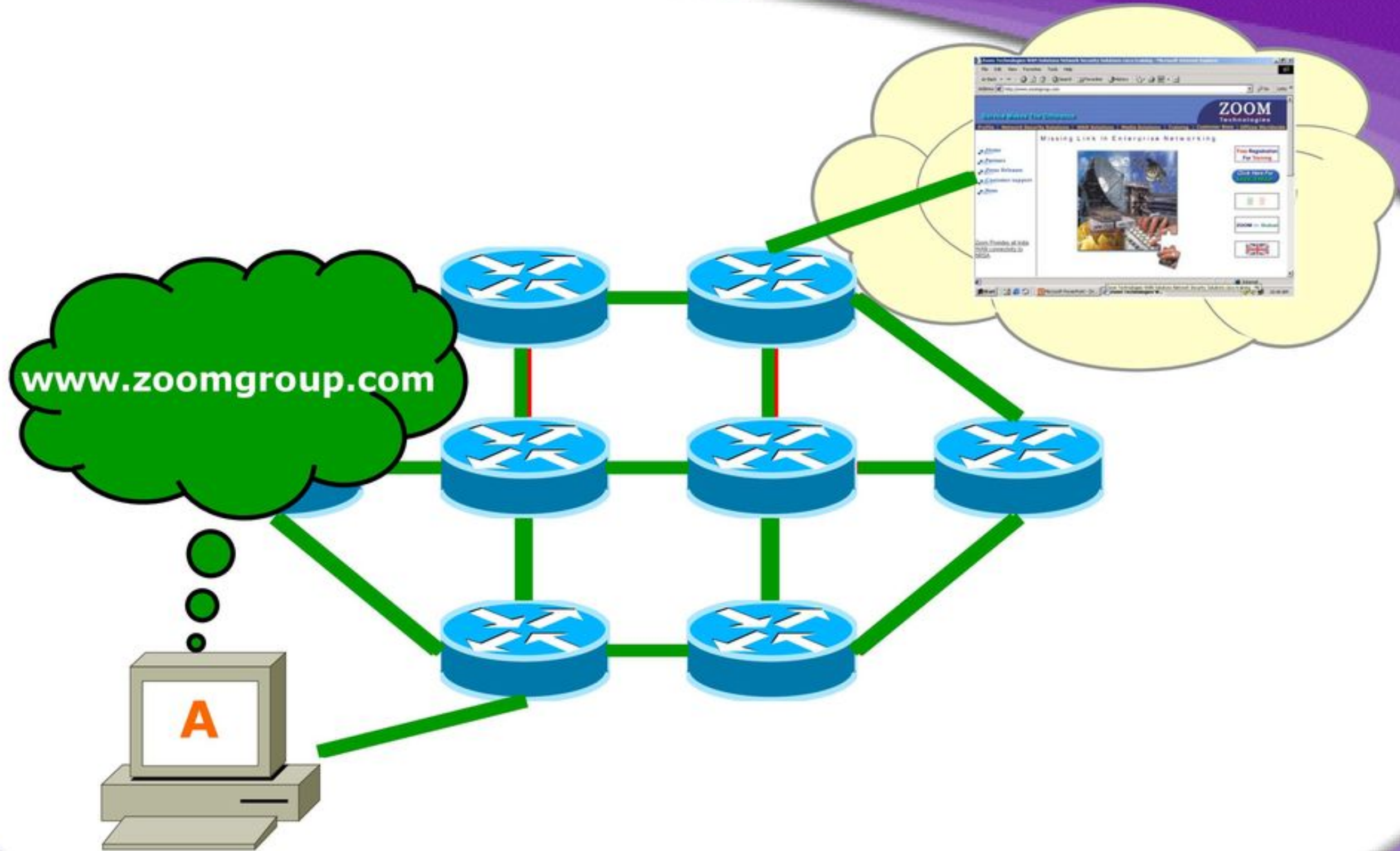


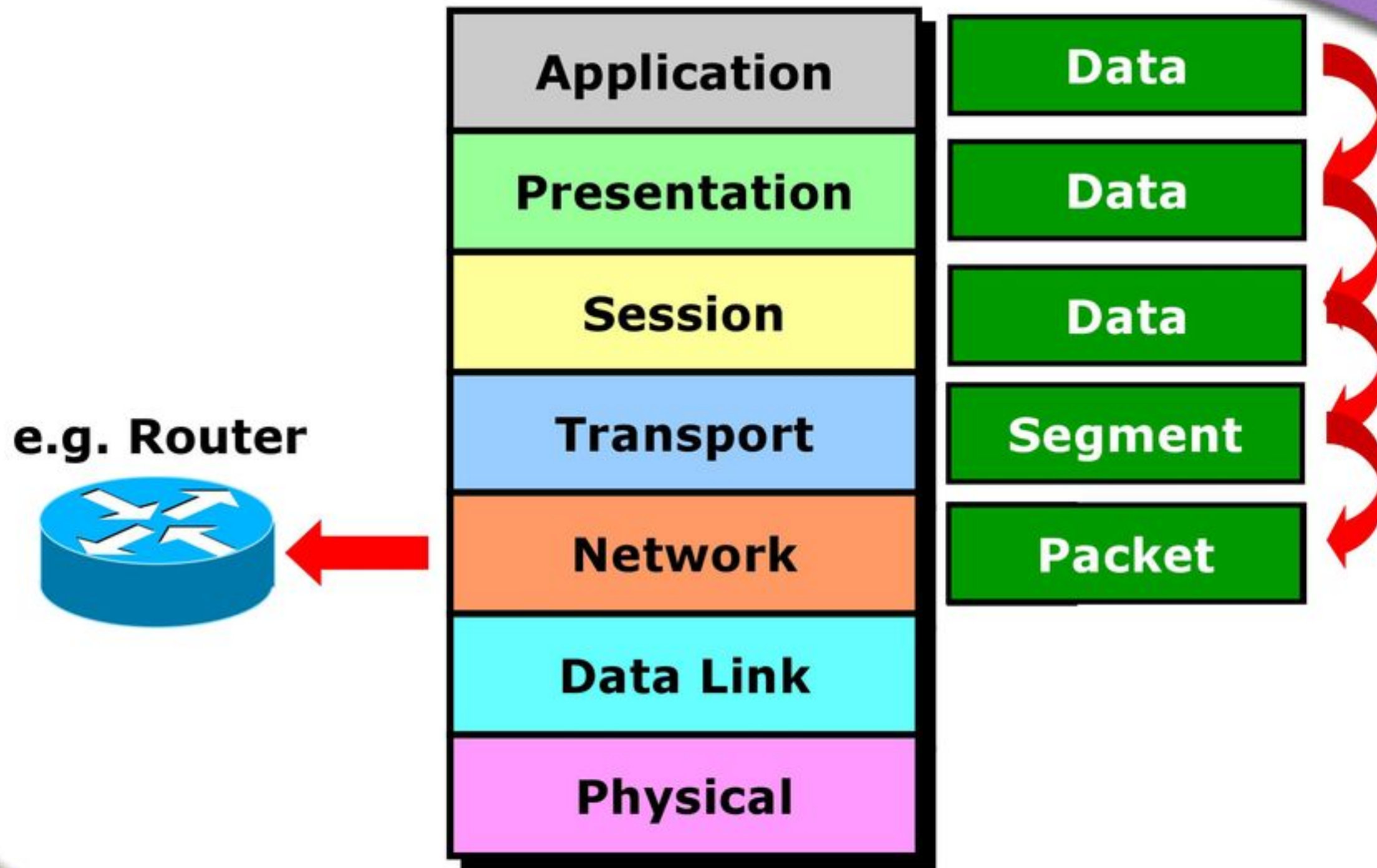
Network Layer is responsible for providing best path for data to reach the destination. Logical Addressing works on this layer. Router is a Network Layer device.

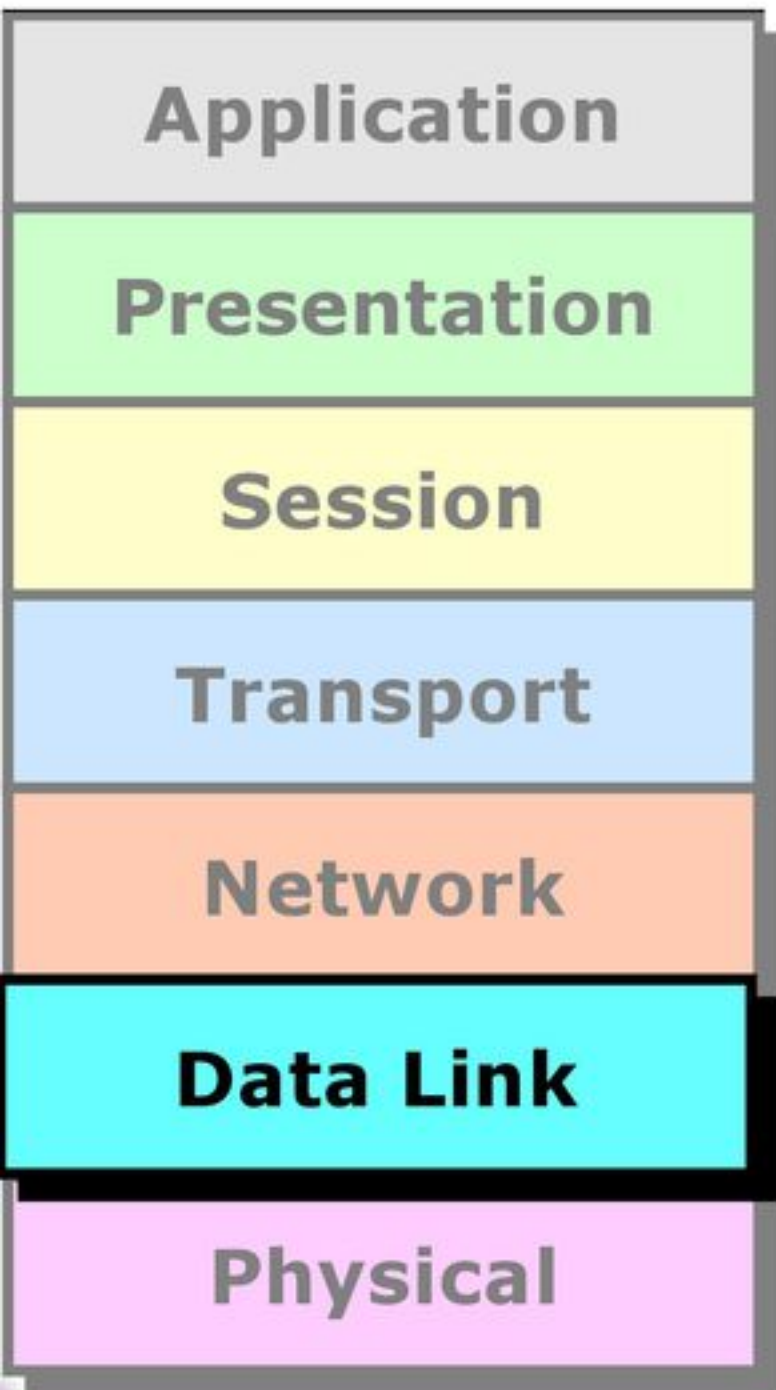
It is divided into two parts

- **Routed Protocols**
e.g. IP, IPX, Apple Talk.
- **Routing Protocols**
e.g. RIP, IGRP, OSPF, EIGRP



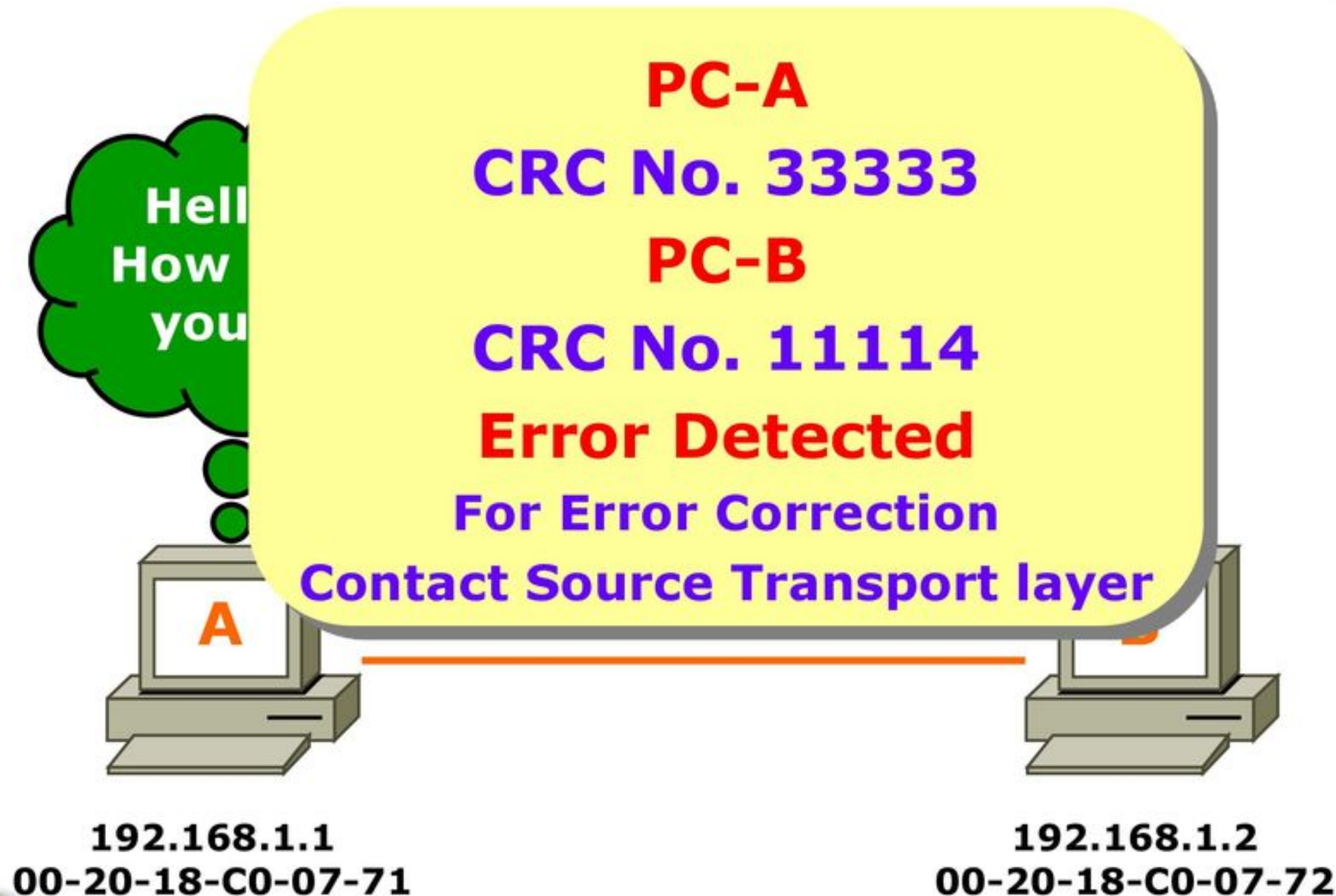






Datalink Layer is divided into two Sub Layers :

- **LLC – Logical Link Control**
It talks about Wan protocols e.g. PPP, HDLC, Frame-relay
- **MAC – Media Access Control**
It talks about Physical Address.
It is a 48 bit address i.e. 12 digit Hexadecimal Number.
It is also responsible for Error Detection
Devices working on Data Link Layer are Switch, Bridge, NIC.



Packet

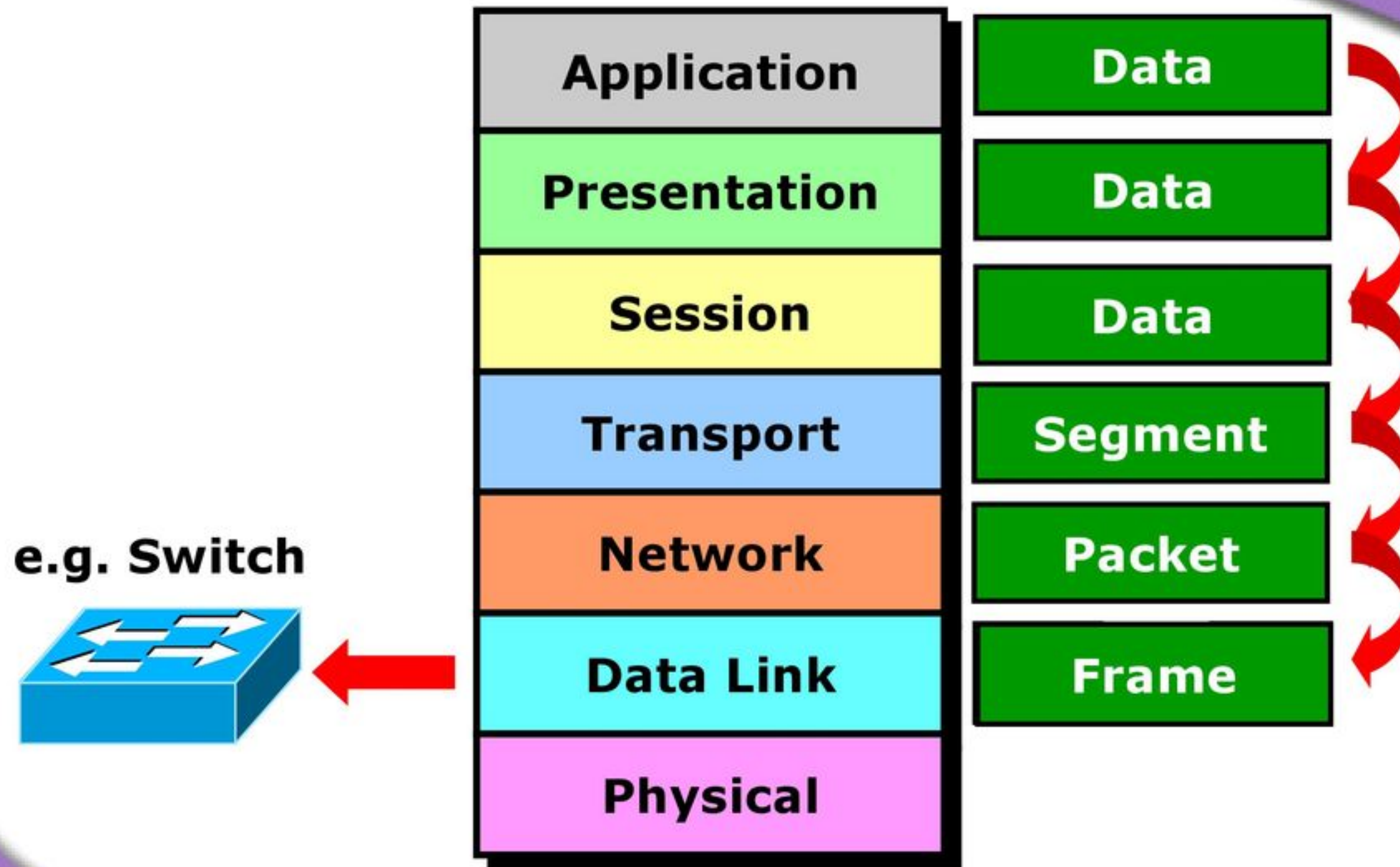
Hello!
How are
you ?

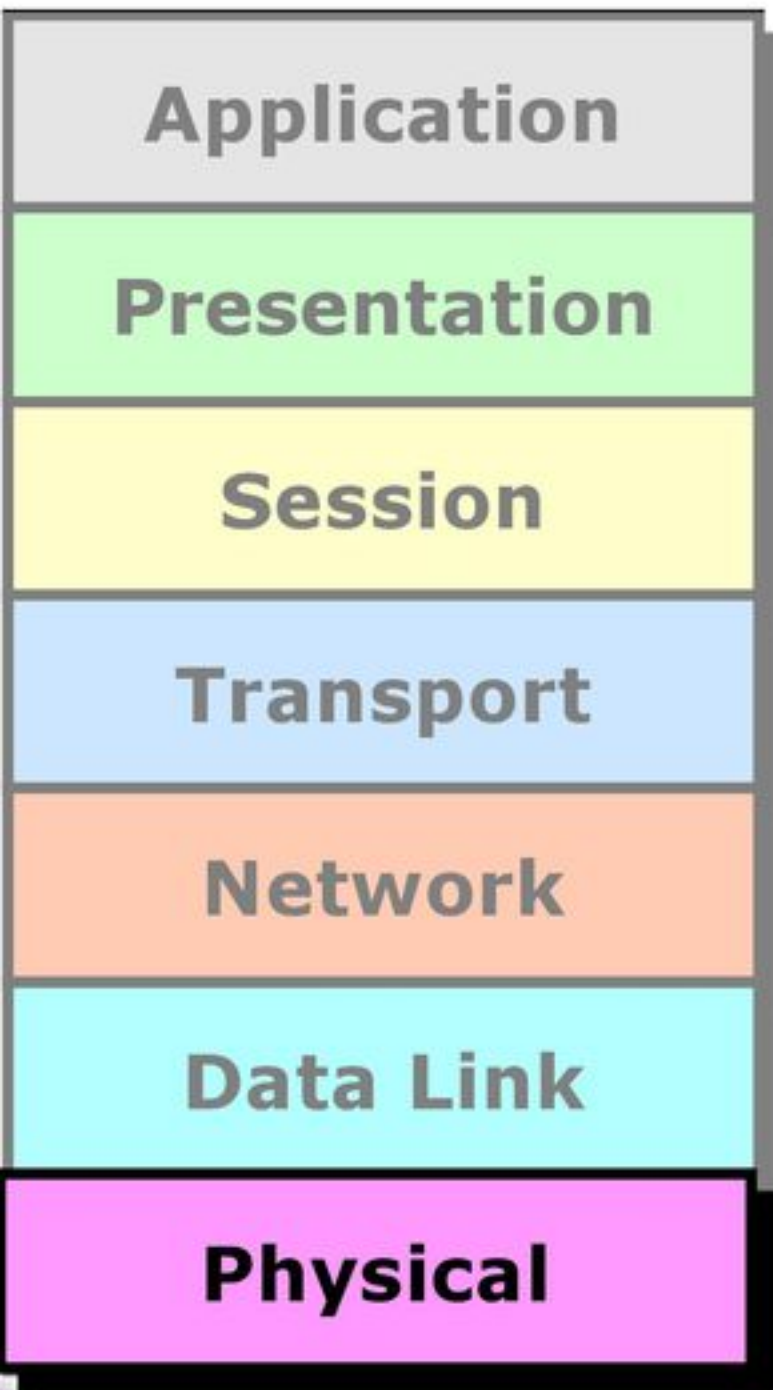


192.168.1.1
00-20-18-C0-07-71



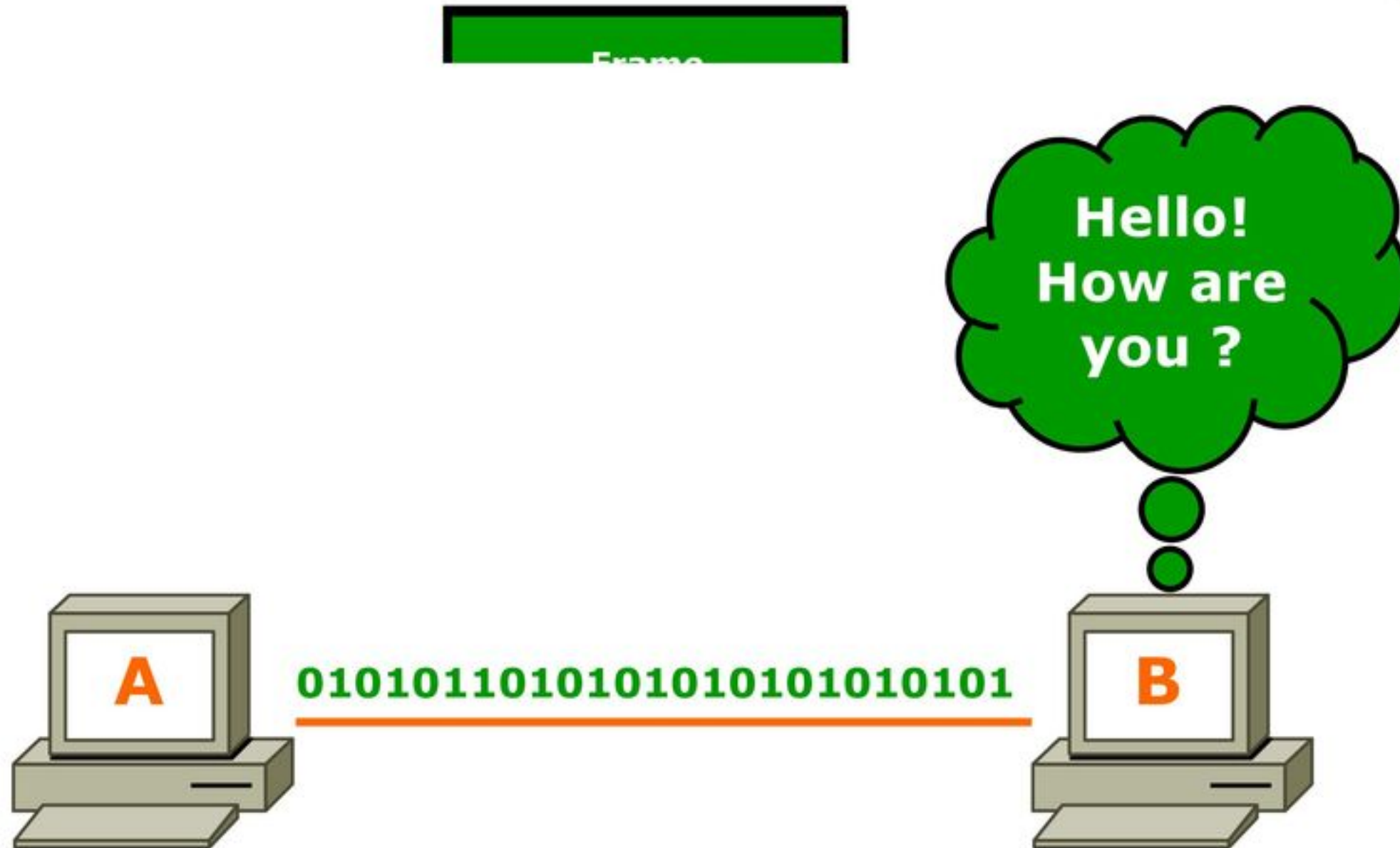
192.168.1.2
00-20-18-C0-07-72

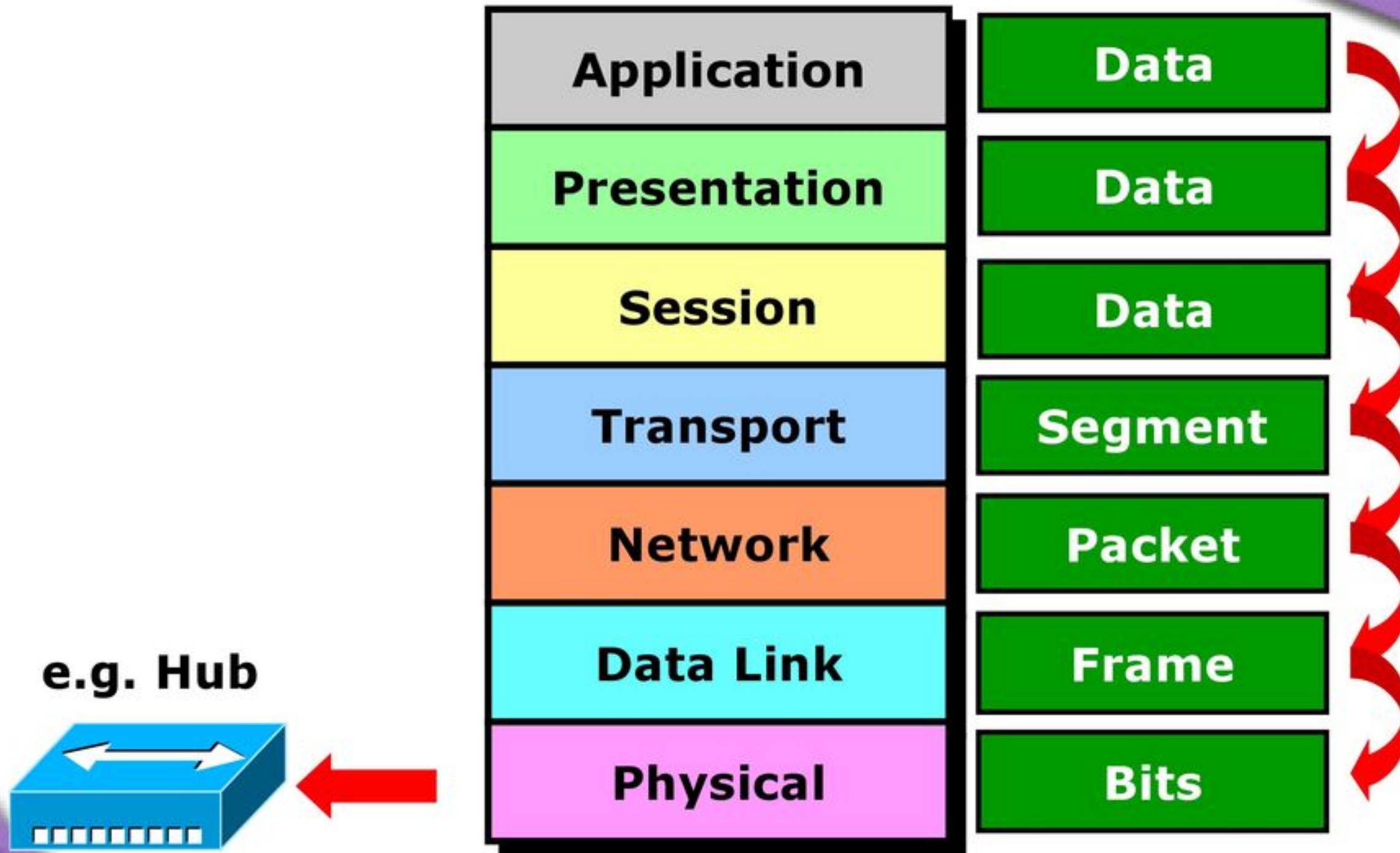




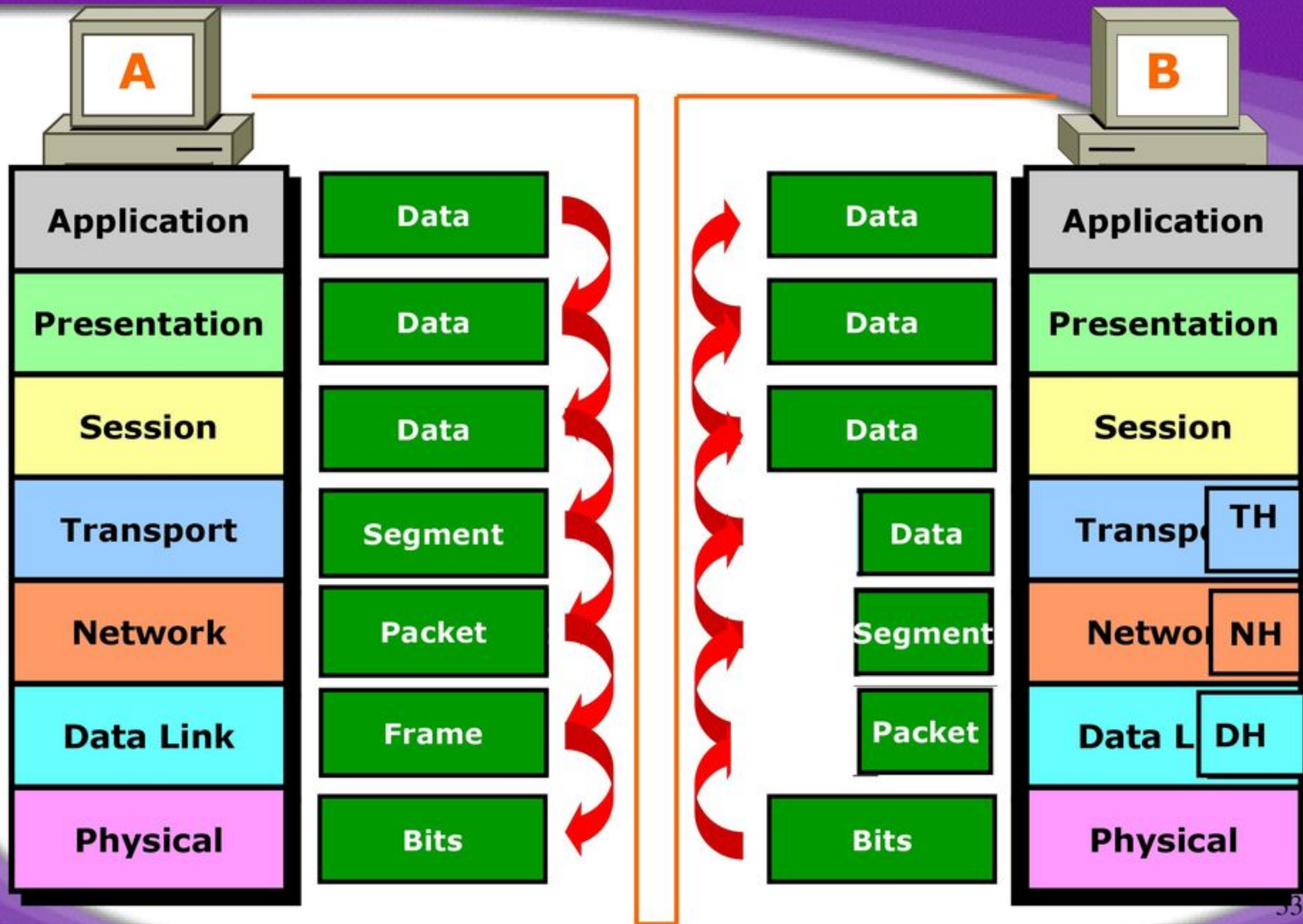
Physical Layer is responsible for electrical, mechanical and procedural checks. Data will be converted into Binary (i.e) 0's & 1's. Data will be in the form of electrical pulses if it is Coaxial or Twisted Pair cable and in the form of Light if it is Fiber Optic Cable.

Devices working at Physical Layer are Hubs, Repeaters, Cables, Modems etc.





Data Encapsulation & De-capsulation



Simulation Scenario

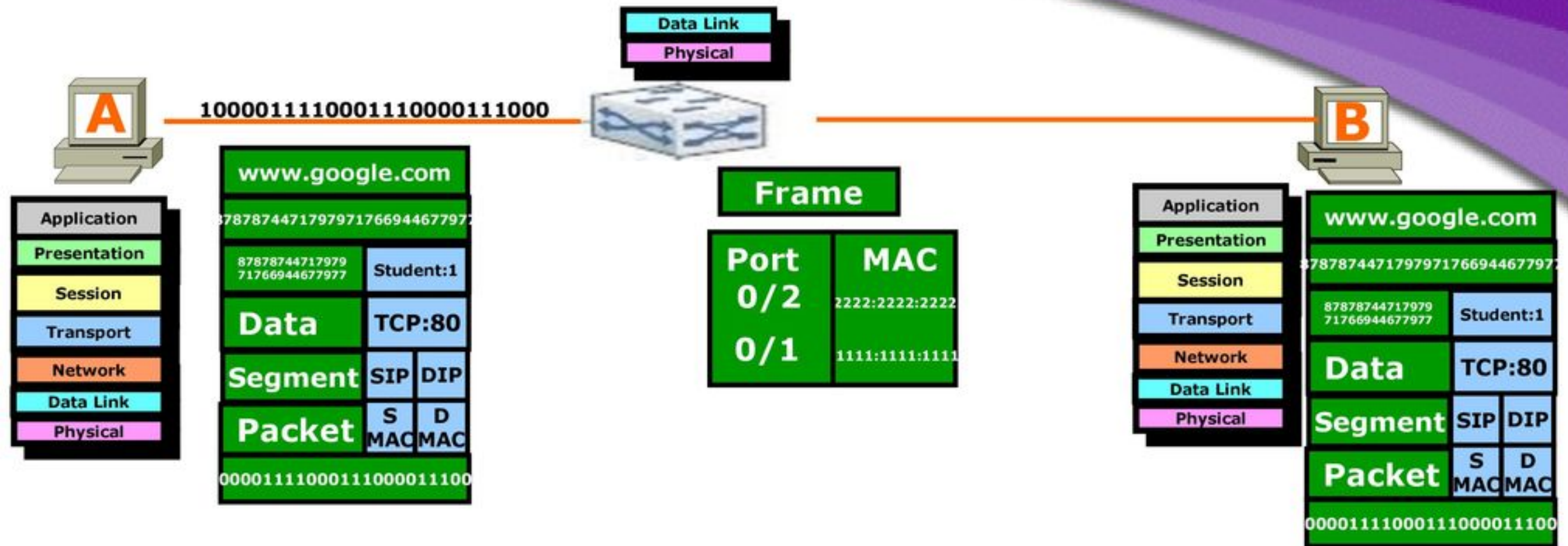


Figure 2.21 *Port addresses*

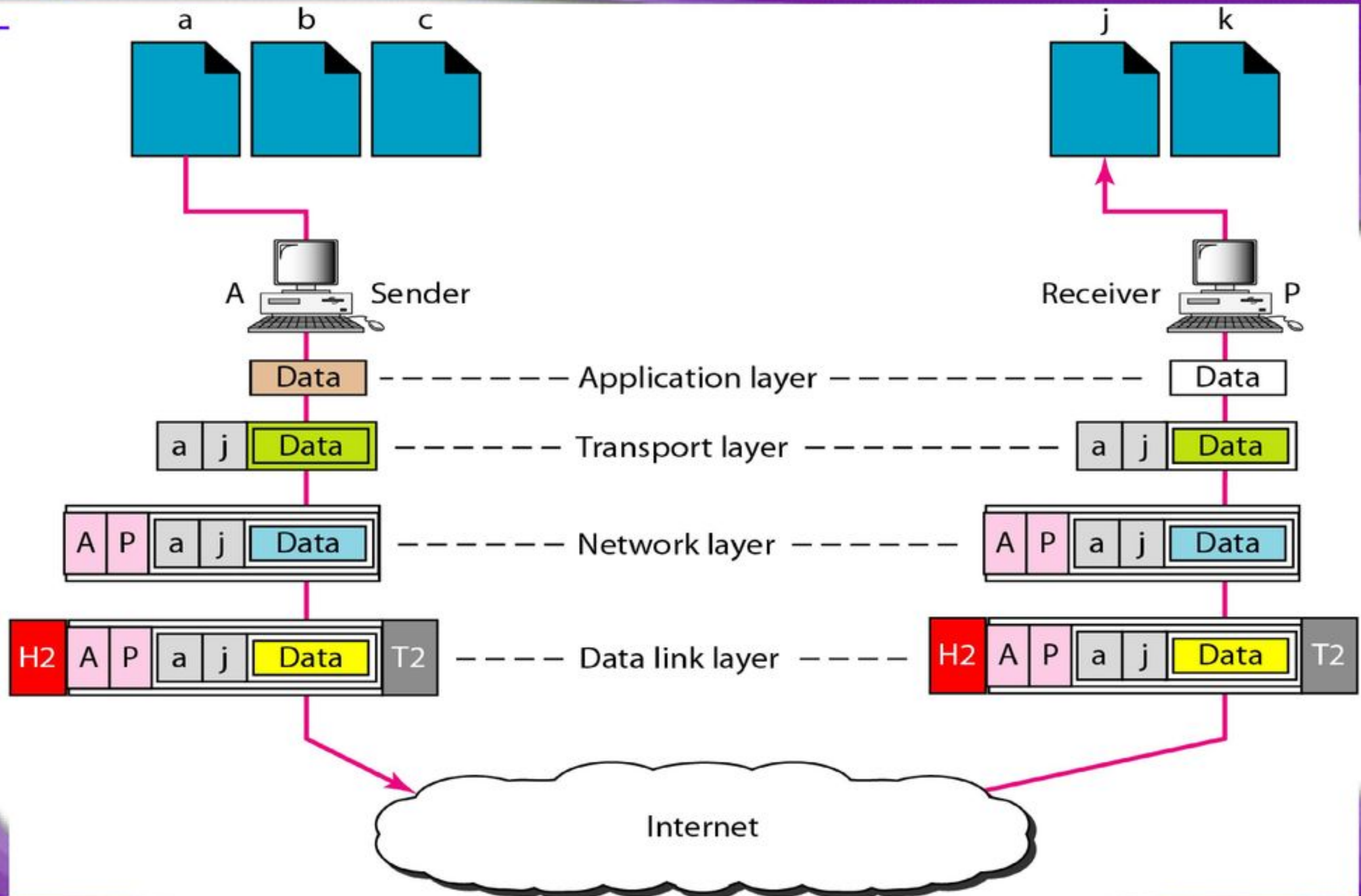
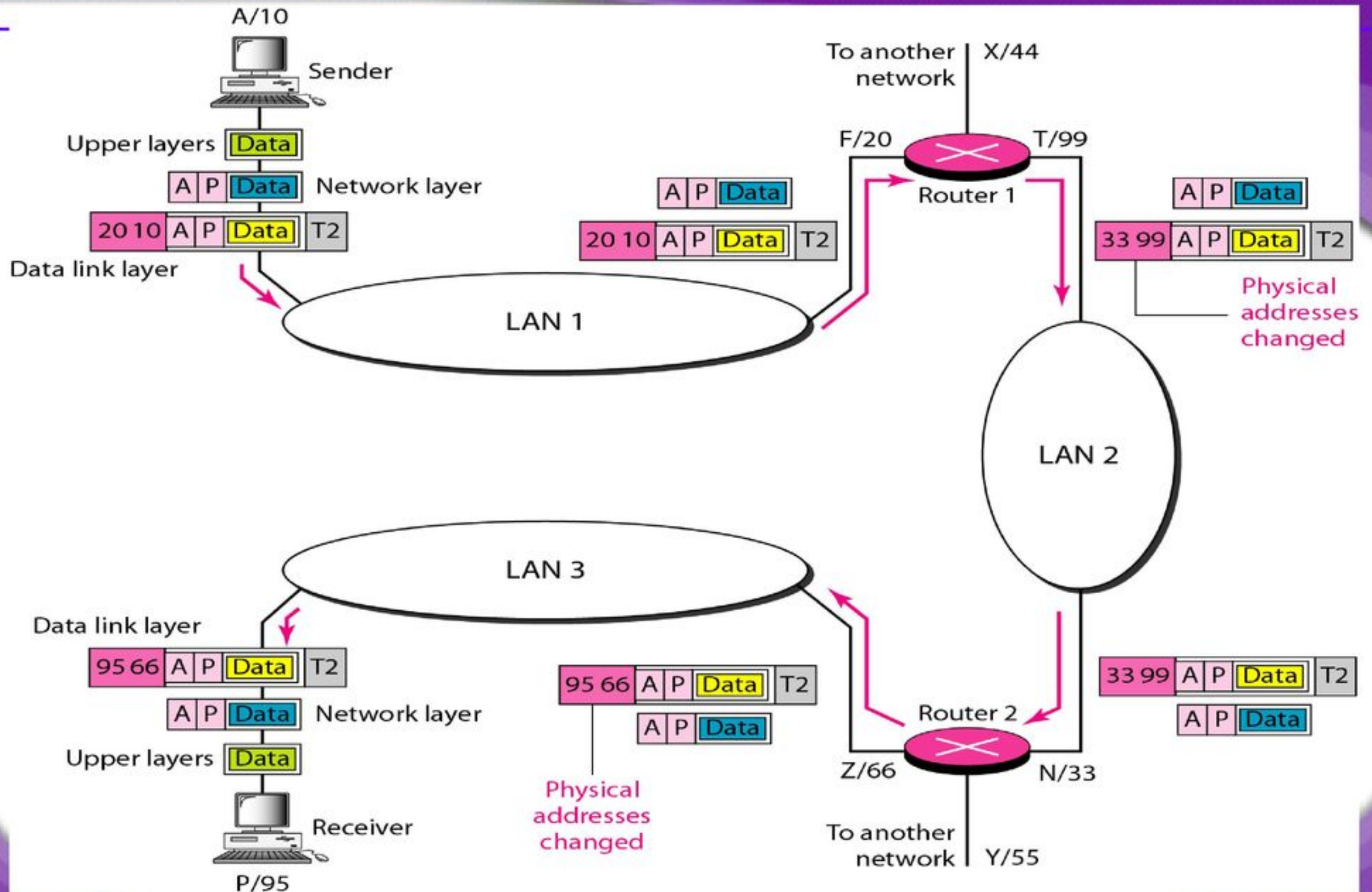
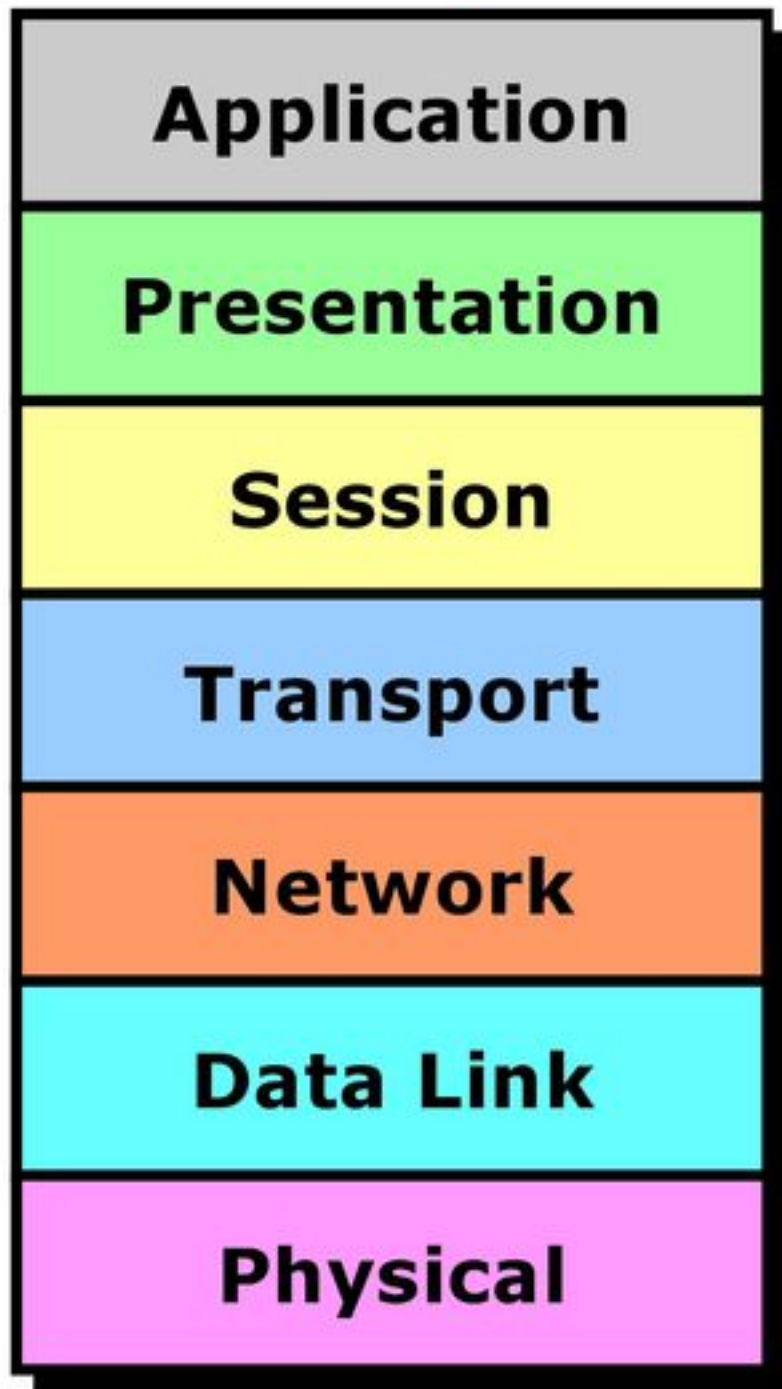


Figure 2.20 *IP addresses*



OSI Layers



TCP/IP Layers

